

Data-Driven Diagnosis For Fuel Injectors Of Diesel Engines In Heavy-Duty Trucks

Emely Björkkvist
Felix Eriksson

Supervisor : Arman Mohammadi, ISY Linköping University
Examiner : Daniel Jung, ISY Linköping University

External supervisor : Victor Petersson, Scania CV AB
Markus Lindblom, Scania CV AB

Upphovsrätt

Detta dokument hålls tillgängligt på Internet - eller dess framtida ersättare - under 25 år från publiceringsdatum under förutsättning att inga extraordinära omständigheter uppstår.

Tillgång till dokumentet innebär tillstånd för var och en att läsa, ladda ner, skriva ut enstaka kopior för enskilt bruk och att använda det oförändrat för ickekommersiell forskning och för undervisning. Överföring av upphovsrätten vid en senare tidpunkt kan inte upphäva detta tillstånd. All annan användning av dokumentet kräver upphovsmannens medgivande. För att garantera äktheten, säkerheten och tillgängligheten finns lösningar av teknisk och administrativ art.

Upphovsmannens ideella rätt innefattar rätt att bli nämnd som upphovsman i den omfattning som god sed kräver vid användning av dokumentet på ovan beskrivna sätt samt skydd mot att dokumentet ändras eller presenteras i sådan form eller i sådant sammanhang som är kränkande för upphovsmannens litterära eller konstnärliga anseende eller egenart.

För ytterligare information om Linköping University Electronic Press se förlagets hemsida <http://www.ep.liu.se/>.

Copyright

The publishers will keep this document online on the Internet - or its possible replacement - for a period of 25 years starting from the date of publication barring exceptional circumstances.

The online availability of the document implies permanent permission for anyone to read, to download, or to print out single copies for his/hers own use and to use it unchanged for non-commercial research and educational purpose. Subsequent transfers of copyright cannot revoke this permission. All other uses of the document are conditional upon the consent of the copyright owner. The publisher has taken technical and administrative measures to assure authenticity, security and accessibility.

According to intellectual property law the author has the right to be mentioned when his/her work is accessed as described above and to be protected against infringement.

For additional information about the Linköping University Electronic Press and its procedures for publication and for assurance of document integrity, please refer to its www home page: <http://www.ep.liu.se/>.

Abstract

The diesel engine in heavy-duty trucks is a complex system with many components working together, and a malfunction in any of these components can impact engine performance and result in increased emissions. Fault detection and diagnosis have therefore become essential in modern vehicles, ensuring optimal performance and compliance with progressively stricter legal requirements. One of the most common faults in a diesel engine is faulty injectors, which can lead to fluctuations in the amount of fuel injected. Detecting these issues is crucial, prompting a growing interest in exploring additional signals beyond the currently used signal to enhance the performance and robustness of diagnosing this fault.

In this work, an investigation was conducted to identify signals that correlate with faulty injectors causing over- and underfueling. It was found that the NO_x , O_2 , and exhaust pressure signals are sensitive to this fault and could potentially serve as additional diagnostic signals. With these signals, two different diagnostic methods were evaluated to assess their effectiveness in detecting injector faults. The methods evaluated were data-driven residuals and Random Forest classifier. The data-driven residuals, when combined with the CUSUM algorithm, demonstrated promising results in detecting faulty injectors. The O_2 signal proved effective in identifying both fault instances, while NO_x and exhaust pressure were more effective at detecting overfueling. The Random Forest classifier also showed good performance in detecting both over- and underfueling. However, it was observed that using a classifier requires more extensive data preprocessing. Two preprocessing methods were employed: integrating previous measurements and calculating statistical measures over a defined time span. Both methods showed promising results, with the latter proving to be the better choice. Additionally, the generalization capabilities of these methods across different operating conditions were evaluated. It was demonstrated that the data-driven residuals yielded better results compared to the classifier, which required training on new cases to perform effectively.

Acknowledgments

This thesis is a collaboration between the Fuel System Control department at Scania CV AB and the Vehicular Systems division in the Department of Electrical Engineering at Linköping University, conducted during the spring of 2024.

We would like to thank our supervisors at Scania, Victor Peterson, and Markus Lindblom, for their help and guidance throughout this thesis project. Your system knowledge and assistance in data collection have been valuable and crucial for the realization of this project.

We would also like to extend our thanks to our supervisor at Linköpings University, Arman Mohammadi, for all his help and support throughout this project. Lastly, we would like to thank our examiner, Daniel Jung, for his additional expertise on the subject.

Linköping, June 2024

Emely Björkkvist

Felix Eriksson

Contents

Abstract	iii
Acknowledgments	iv
Contents	v
1 Introduction	1
1.1 Background	1
1.2 Problem Formulation	2
1.3 Delimitations and Assumptions	3
1.4 Related Research	3
2 System Description	5
2.1 System Overview	5
2.2 Fuel Injection System	6
2.3 Data Collection	6
3 Identification of Informative Target Signals for Diagnosis of Injector Faults	8
3.1 Analysis of System and Literature	8
3.2 Analysis of Experimental Data Using Variable Selection	9
3.2.1 Absolute Difference	9
3.2.2 Feature Importance Using Random Forest	11
3.3 Result of Signal Identification Process	13
4 Development of Residual Based Diagnosis System	15
4.1 Residual Generator	15
4.2 Hypothesis Test	15
4.3 CUSUM Test Quantity	16
4.4 Choice of Model Structure	17
4.5 Construction of Data-Driven Residuals	18
4.6 Input Selection for Each Target Signal	20
5 Development of Random Forest Classifier	22
5.1 Data Preparation	22
5.2 Implementation	24
5.3 Evaluation	24
5.4 Data Visualization	25
6 Results	26
6.1 Data-Driven Residuals	26
6.2 Random Forest Classifier	31
6.2.1 Overfueling	31
6.2.2 Underfueling	33

6.2.3	Data Visualization	35
6.3	Operating Point Assessment	36
6.3.1	Data-Driven Residuals	36
6.3.2	Random Forest Classifier	37
7	Discussion	43
7.1	Data Collection	43
7.2	Identification of Informative Target Signals for Diagnosis of Injector Faults . . .	44
7.3	Data-Driven Residuals	44
7.3.1	Evaluation of Model Performance and Design	44
7.3.2	Evaluation of Diagnosis and Residual Performance	45
7.4	Classification	46
7.4.1	Evaluation of Performance and Input Method	46
7.4.2	Further Refinement for Utilizing Classifiers in Diagnostic Purposes . . .	47
7.5	Operating Point Assessment	47
7.5.1	Data-Driven Residuals	47
7.5.2	Classification	48
7.6	Discussion of Diagnostic Methods	49
7.6.1	Data-Driven Residuals	49
7.6.2	Classification	50
7.6.3	Compilation and Comparison of the Diagnosis Methods	50
7.7	The Work in a Wider Context	52
8	Conclusion and Future Work	53
8.1	Future Work	54
	Bibliography	55

Abbreviations and Nomenclature

Abbreviations

ARX Autoregressive Exogenous

CUSUM Cumulative Sum

EMS Engine Management System

FCN Function

FI Feature Importance

HPP High-Pressure Pump

LPP Low-Pressure Pump

ML Machine Learning

NRMSE Normalized Root Mean Square Error

NARX Nonlinear Autoregressive Exogenous

RF Random Forest

SS State-Space

t-SNE t-Distributed Stochastic Neighbor Embedding

Nomenclature

ν CUSUM tuning parameter

O_2 Oxygen

NO_x Nitrogen Oxide

P_{exh} Exhaust Pressure

CO Carbon Monoxide

CO_2 Carbon Dioxide

r Residual

T Test Quantity

J Threshold

y Output

$\hat{y}$ Predicted Output

u Input

H Hypothesis

TS Sample Time

Chapter 1

Introduction

1.1 Background

Commercial vehicles like trucks play a critical role in transportation networks. When these vehicles experience malfunctions, it can disrupt the entire logistics system, leading to considerable economic losses and environmental consequences. The environmental impact of vehicle emissions has gained increased attention in recent years, leading to progressively stricter legal requirements for emissions control [1]. The industry must adapt to these regulations to meet sustainability goals. Fault detection and diagnosis is essential in modern vehicles to reduce emissions and ensure optimal performance by detecting when something is not working as it should. Additionally, effective diagnostics also support efficient operations. By detecting faults early and isolating them, downtime is minimized and costs are reduced.

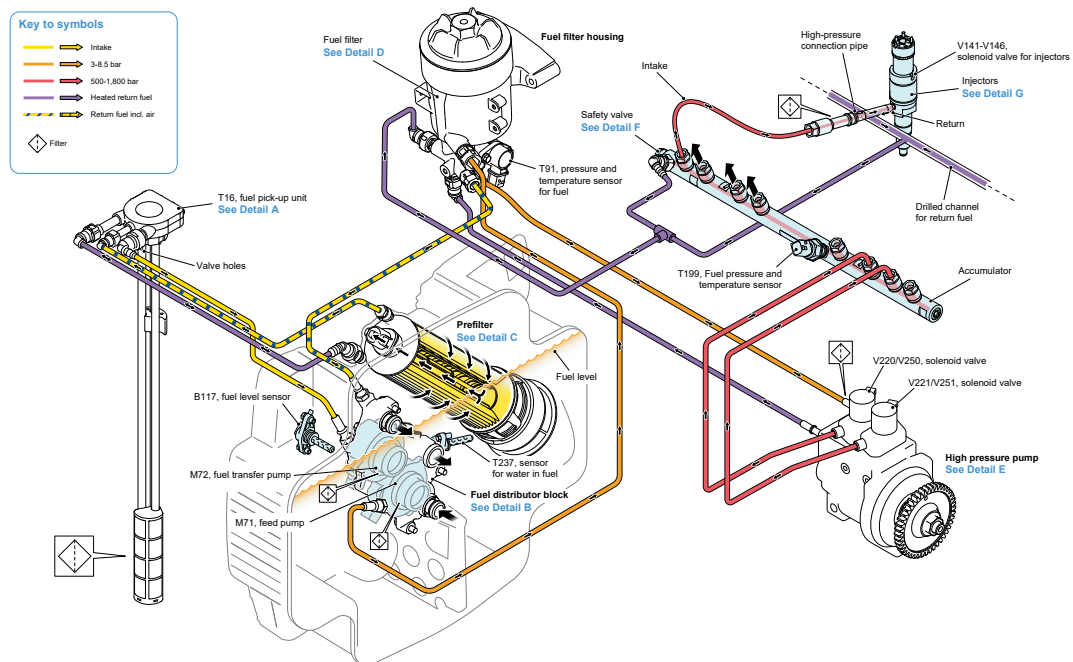


Figure 1.1: System overview of the Scania truck fuel system. ©Scania CV AB 2024, Reprinted with permission.

The diesel engine is a complex system with many components optimized to work together for optimal performance and low emissions. Figure 1.1 illustrates the fuel injection system, a sub-system responsible for regulating and delivering fuel to the engine's cylinders. Faulty fuel injectors are among the most common issues in diesel engines, and if an injector does not behave as expected, it can lead to excessive or insufficient injection quantities, known as over- and underfueling.

- **Overfueling:** Injector delivers more fuel than expected.
- **Underfueling:** Injector delivers less fuel than expected.

This can negatively affect both engine performance and increase tailpipe emissions, emphasizing the importance of diagnosing and monitoring injector characteristics as a critical part of the control system [2].

Research on the topic of fuel injector diagnosis has been conducted using both system-specific insights [3] and data-driven approaches [4]. Currently, Scania trucks use engine angular velocity as an input to diagnose over- and underfueling, but there is a growing interest in exploring additional signals that could improve diagnostic accuracy and reliability. This leads to the question of which existing signals in the truck can offer more insights into fuel-related faults. Sensors monitoring exhaust emissions, for example, have been investigated for their ability to provide information about combustion efficiency [5]. Furthermore, incorporating multiple sources that indicate abnormal behavior will result in more accurate diagnoses. By combining this knowledge with additional data-driven methods, diagnostic accuracy can be enhanced. This could potentially decrease the number of false alarms and missed detections, thereby enhancing engine performance and reducing emissions.

1.2 Problem Formulation

The current research presents a gap concerning the existence of additional signals in a heavy-duty truck that correlate with over- and underfueling. This thesis aims to fill this gap by exploring potential correlations within the system.

The primary objective of this thesis is therefore to identify signals that correlate with over- and underfueling. This will be done by gaining an overview of the system to see causality. By integrating existing research findings in literature and validating them against the collected data, conclusions can be drawn based on multiple factors. The identified signals will then be used to evaluate different diagnostic methods. Lastly, the performance of the diagnostic methods will be evaluated across different operating points. The evaluation will be based on experimental data from both fault-free and faulty scenarios.

By addressing these objectives, this thesis aims to contribute to the development of more effective diagnostic methods. The specific questions that this thesis intends to address are:

1. Identify additional signals for diagnosing injectors with over- and underfueling.
2. Evaluate different diagnostic methods and propose a diagnostic system designed to detect both over- and underfueling.
3. Evaluate how different driving conditions and operating points affect the detection and diagnosis of over- and underfueling.

1.3 Delimitations and Assumptions

Because of limited time and resources, this project has the following delimitations to ensure the right focus and that the time is evenly distributed during the project. All injectors will be affected by the injected fault during data collection, which means that faults in one injector only will not be evaluated. Other faults in the system, besides over- and underfueling, will not be considered in this project.

The following two assumptions will be made throughout this thesis:

- The truck that the data is collected from is working correctly and has no unknown faults.
- The rigged faults in the injectors represent real faults in the injectors.

1.4 Related Research

Several studies have employed a model-based approach for fault detection. The authors in [6] investigate enhancing diesel engine diagnosis through a model-based method using signal and process models. Similarly, system models are applied in both [3] and [7] to develop a diagnosis system for common-rail injection systems and the Selective Catalytic Reduction (SCR) system, respectively. However, in recent years, data-driven approaches have become increasingly popular for developing diagnosis systems. In [4], the author evaluates different machine learning (ML) algorithms for detecting faults related to the fuel injector, using the engine angular velocity as input. The authors in [8] followed a similar approach, investigating three ML methods for diagnosing the fuel injection systems of a two-stroke diesel engine.

Machine learning is a powerful method that allows computers to learn patterns and behaviors in data without the need to program them to do certain things. In fault diagnosis, ML is gaining popularity for its effectiveness. While model-based methods are common in fault diagnosis, they can be limited by the complexity of accurately modeling the system. Using ML algorithms offers a promising alternative. For instance, [9] conducted a literature review on ML techniques in mechanical fault diagnosis, highlighting the increasing number of studies on this topic in the manufacturing industry. They also state that further research is needed on this topic to address the challenges posed by real-world scenarios.

In [10], a hybrid method is utilized for a proposed diagnosis system. The combination of model-based and data-driven approaches contributes to an algorithm for selecting residuals. This algorithm aims to identify a set of residual generators that maximizes the performance of fault detection and isolation. To validate the method, it is applied to a diagnosis system that monitors the air path through an internal combustion engine. In another report [11], researchers have proposed an alternative hybrid method. They introduced a structural model of the fuel injection system by developing a mathematical model through basic component models. The methodology includes employing structural analysis to extract redundant sensor sets and establish causal relationships between signals. From this, a set of data-driven residual generators are derived, which are then assessed using experimental data from fault-free and faulty scenarios.

Employing a structural analysis when designing a diagnosis system has been widely used in prior studies. For instance, in the previously mentioned article [7], authors used this method when developing their model-based diagnosis. When constructing this type of analysis, the Fault Diagnosis Toolbox (FDT) [12], proves to be a valuable resource. Noteworthy features of the toolbox include support for the structural analysis of large-scale dynamic models, residual generator analysis, and test selection.

To conduct a structural analysis, a system model is essential. For instance, a simplified model of the Selective Catalytic Reduction (SCR) system located in the aftertreatment has been developed in [7], considering only the high-pressure side of the urea dosing system. The SCR model excludes the vent in the tank and the pressure relief valve in the pump since these do not affect the nominal system behavior. In another report [11], researchers have introduced a structural model of the fuel injection system by developing a mathematical model through basic component models. Consistent with previously referenced articles, their methodology includes employing structural analysis to extract redundant sensor sets and establish causal relationships between signals. The primary focus lies in crafting data-driven residual generators, which are then assessed using experimental data from fault-free and faulty scenarios.

Another article that develops a model for detecting faults in the fuel injection system is [13]. The author of that work also considers the combustion process and the crankshaft drive. They generate three residuals each from engine speed and oxygen (O_2) evaluations. One particularly interesting residual from this article is created by comparing the nominal injection mass with the reconstructed mass. The latter is calculated from the measured O_2 concentration and the aspirated air mass over one working cycle. There are numerous models of diesel engines in the literature. For this project, simple models are preferred. An article that presents such a model is [14]. They present a static non-linear model for components like the throttle, mass flow, engine torque, engine out temperature, and turbocharger, which is of interest in this project.

In [8] they elaborate on the engine performance and that the exhaust gas of a diesel engine contains a substantial amount of information regarding the combustion efficiency. They highlight the volume of gas components like Hydrocarbons, Carbon Oxides, and Nitrogen Oxide (NO_x). The authors in [15] propose a method for diagnosing engine faults using sensor data flow and exhaust emission information, which resulted in high diagnostic accuracy. The authors in [5] made a similar finding where they used a composition of the exhaust gas as a diagnostic signal and presented a multidimensional diagnosis tool. The authors proved with experiments that a delayed fuel injection in one cylinder corresponded to an increase in Carbon Dioxide (CO_2) and a decrease in NO_x emissions. The Carbon Monoxide (CO) emission increased only in high engine loads.

Many signals can be taken into consideration when looking at the fuel, combustion, and exhaust system, therefore it is needed to have a systematic way to do a first selection on which signals can be interesting to look at. In [16], Hultgren used feature selection methods to identify signals that correlated the most with changes in fuel composition. He used a feature selection method based on SHapley Additive exPlanations (SHAP) and applied it to an expert network to find the features that correlated the most with change in fuel composition.

The mentioned literature demonstrates that investigations related to this topic have been conducted using various methods. These investigations have identified signals that contain information relevant to this fault. This work aims to extend this research by enhancing the understanding of which signals contain information about this fault and how these signals can be used effectively and robustly. Hopefully, this will advance the knowledge and efficiency of diagnosing faulty injectors causing over- and underfueling.

Chapter 2

System Description

This chapter provides a general overview of the fuel system and the signals used in this project. It also describes the data collection procedure.

2.1 System Overview

Figure 2.1 provides a simplified schematic of the diesel engine, highlighting essential components and their connections. It shows some investigated signals and their placements, with colored signals being the most relevant. Note that the figure is not fully representative, as not all components are included, and some sensor placements are estimations due to the simplification. The system sucks air in through the air cleaner and compresses it with the compressor to provide additional air to the engine. The compressed air proceeds to enter the intake manifold, where it is subsequently pulled into each cylinder. The fuel system consists of filters and two pumps. The Low-Pressure Pump (LPP) transfers fuel from the tank to the High-Pressure Pump (HPP), which pressurizes the fuel in the common rail. The injectors then spray fuel into the cylinder, where it mixes with the air. This mixture combusts in each cylinder during different cycles, generating engine torque. The remaining combustion byproducts exit through the exhaust manifold, driving the turbine, which in turn powers the compressor. The wastegate controls exhaust gas bypass past the turbine for compression regulation. Finally, the exhaust gas passes through aftertreatment for cleaning.

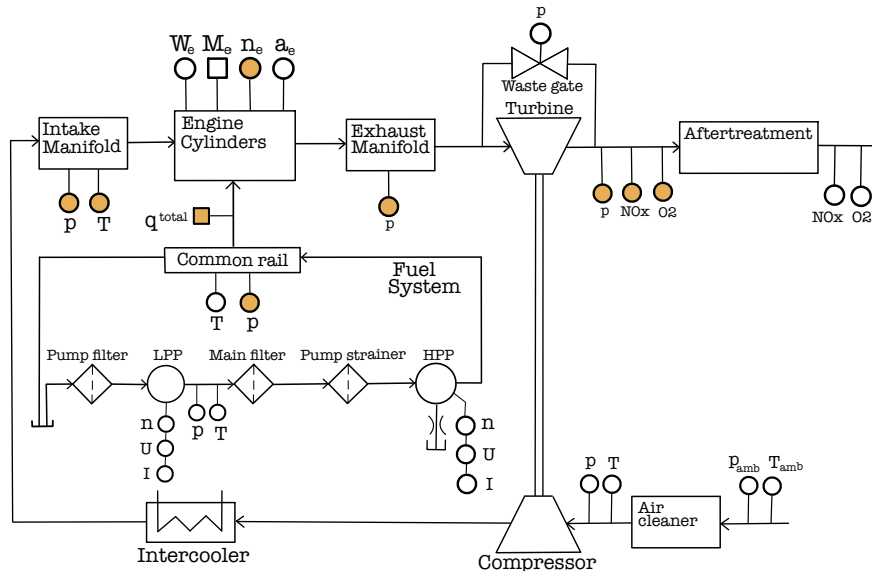


Figure 2.1: Simplified overview of the diesel engine, showcasing key signals investigated in this project. Colored signals represent the most relevant ones. Note that, due to simplification, not all components are included and some sensor placements are estimates.

2.2 Fuel Injection System

The fuel injectors play a critical role in ensuring the correct amount of fuel is delivered to the engine. The quantity of fuel injected depends on the pressure differential between the combustion chamber and the common rail, as well as the injector's opening duration. The Engine Management System (EMS) controls the opening duration to ensure the desired amount of fuel is injected into the combustion chamber.

Faulty injectors are one of the most common faults in diesel engines, with clogging being one of the causes. During fuel injector operation, residues from carbonized fuel and microscopic foreign objects can accumulate inside the nozzle, leading to blockages that result in irregular injection patterns, a weak spray, and uneven fuel distribution. These symptoms tend to worsen gradually if the injectors are not replaced, potentially causing them to fail in delivering fuel to the cylinder. Problems such as faulty solenoids, overheating, short circuits, and rust can impact the proper opening and closing of valves. If the valve is unable to close properly, it can result in fuel leakage and uncontrolled emissions or combustion. Furthermore, the occurrence of stiction can introduce delays in valve movement, further complicating the functionality of fuel injectors [17].

2.3 Data Collection

The data was collected at Scania in Södertälje at their test facility, using a heavy-duty truck with a diesel engine. The same truck was used during all tests. The data consists of a total of 112 signals, with a sampling frequency of 10 Hz. To obtain the faulty data, an actual fault was not implemented in the system. Instead, the truck was modified to simulate faulty injectors. This modification involved overriding the EMS, causing the injectors to inject more or less fuel than the system desired over a set period of time. The modification generates a constant fault measured in milligrams per stroke [mg/st] as illustrated in Figure 2.2, where the black dotted lines indicate the stop and reset of a new dataset with a new fault size.

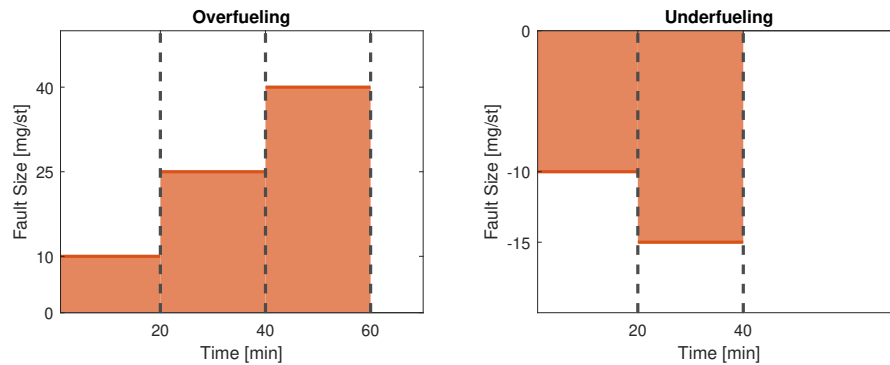


Figure 2.2: Illustration showing the representation of constant faults in the data for over-fueling and underfueling. The dotted line indicates where the data collection stopped and restarted with a new fault size.

The collected data include four different cases outlined below. The no-fault (NF) case served as the baseline for the truck's nominal behavior, whereas fault cases F1 and F2 depicted its behavior during over- and underfueling, respectively. Additionally, the fault-free operation with a load (NF Load) was employed for assessing the diagnostic system's generalization across another circumstance. The load used was an attached trailer with a weight of 17

tonnes. Initially, the truck was driven to complete the warm-up phase before data collection began. In total, approximately 20 minutes of data were collected per fault. The fault scenarios used in this study are as follows:

- No-fault (NF)
- No-fault with load (NF Load)
- Overfueling (F1)
- Underfueling (F2)

Data was gathered from two distinct driving scenarios: normal driving and aggressive driving with a load. Normal driving aimed to replicate the truck's typical usage in everyday situations. During this test, the truck drove on a closed-loop track, ensuring consistent driving conditions. The velocity remained stable at approximately 80 km/h on straight road segments, while slowing down on turns and inclines.

The second driving condition, aggressive driving with load, was designed to gather faulty data for evaluating the system's behavior across various operating conditions and its impact on the diagnostic system. This test was conducted on a different test track, simulating urban environments, and involved more vigorous maneuvering of the truck. The driving entailed frequent starts and stops, substantial acceleration, and occasional manual gear shifts. The datasets collected during the two driving conditions are presented in Table 2.1. Only faulty data was collected during the aggressive driving condition.

Table 2.1: List of faults and corresponding fault sizes collected for each driving condition.

Normal Driving		Aggressive Driving	
Fault Class	Fault Size	Fault Class	Fault Size
F1	10 mg	F1	10 mg
F1	25 mg	F2	10 mg
F1	40 mg		
F2	10 mg		
F2	15 mg		
NF	[-]		
NF Load	[-]		

Chapter 3

Identification of Informative Target Signals for Diagnosis of Injector Faults

In this chapter, the necessary theory and method for identifying signals that correlate with over- and underfueling will be presented. The goal is to find informative target signals that indicate when a fault is occurring. First, the system and literature will be analyzed to list relevant signals identified in the literature. Two variable selection methods will then be used to verify and complement the findings. Based on the accumulated knowledge, a decision will be made, and the identified target signals will later be used in the evaluation of diagnostic methods.

3.1 Analysis of System and Literature

Understanding how components in the system are interconnected and interact provides insight into which components are likely affected by over- and underfueling, thus hinting at potential correlations with certain signals. For instance, the system description in Figure 2.1 suggests that post-combustion signals are more likely affected by injector faults due to their downstream placement.

Fuel injectors are calibrated to deliver a specific fuel quantity, and any deviation will disrupt this balance. Several studies have investigated how different fuel injection strategies influence combustion performance and NO_x emissions. For instance, [18] found that delaying the injection timing, which results in less fuel being injected before ignition, led to lower peak pressure and combustion temperature, thereby reducing NO_x emissions. Conversely, advancing the start of injection, which results in more fuel being injected before ignition, led to higher combustion temperatures and increased NO_x emissions. Although this study did not alter the specific fuel timing, it provides valuable insights into the effects of injecting more or less fuel before ignition, which is likely to occur if an injector is faulty. The quantities affected by this, as demonstrated in [18], were the same ones commonly used as signals in other studies investigating fault diagnosis for faulty injectors. In [15], ten signals, including the O_2 sensor, NO_x sensor, mass airflow sensor, coolant temperature sensor, throttle position sensor, crankshaft position sensor, and intake air temperature sensor, were used to diagnose engine faults, including injector faults. The NO_x and O_2 signals were also used in [5], along with CO , CO_2 , and exhaust gas temperature signals, for diagnosing fuel injector faults. Furthermore, [8] used cylinder pressure to diagnose fuel system faults, including injector faults. Considering these studies, it is reasonable to expect that exhaust gas signals, as well as signals measuring pressure and temperature near the combustion area, will be impacted by the fault.

To accurately identify fault-induced variations with the identified target signals, minimizing the influence of external factors is important. This involves prioritizing signals less sensitive to such factors while ensuring they provide maximum information on injector faults. For instance, signals near an actuator might reflect its operation rather than the fault itself, making it difficult to attribute variations directly to the faults. By focusing on signals less likely to be affected by external variables, the risk of selecting signals that exhibit sensitivity between no-fault and fault conditions due to reasons unrelated to the fault itself is reduced.

3.2 Analysis of Experimental Data Using Variable Selection

In this step, it became crucial to filter out irrelevant signals before applying variable selection. This step was necessary because the chosen methods focused solely on the differences in the data between no-fault and faulty conditions to reduce the risk of these signals showing a high level of correlation, even though any observed differences were clearly influenced by other factors, such as variations in test conditions.

The process began by identifying all constant signals. This was done by plotting all signals together to visualize their behavior across the dataset. This step reduced the total number of signals from 112 to 69 by eliminating those that exhibited no variation and remained constant throughout the dataset. Unreasonable signals and those significantly distant from the fault or influenced by experimental conditions were then removed. This involved using the system description and plotting the signals for each fault size. Additionally, redundant signals conveying similar information under different names were also removed. This step brought down the total count of signals from 69 to just 23. These steps ensured that the datasets only contained relevant and meaningful signals for further analysis.

The next step involved identifying which of the remaining signals were most affected by over- and underfueling. To achieve this, the absolute difference was used to determine which signals exhibited the greatest separation between fault-free and faulty conditions, while Feature Importance (FI) in Random Forest (RF) was utilized to determine which signals are most influential in a learning model. This approach was meant to serve as a secondary input, grounded in experimental data, to validate the signals identified in the literature as being informative for diagnosing faulty injectors.

3.2.1 Absolute Difference

To ensure a fair comparison, each signal was initially normalized based on its behavior during no-fault conditions. This normalization process utilized maximum normalization, defined as

$$y_{x,\text{norm}} = \frac{y_x}{\max(|y_{x,\text{nf}}|)} \quad (3.1)$$

where each signal, denoted as y_x , was divided by the maximum absolute value of the corresponding no-fault signal, $y_{x,\text{nf}}$, to obtain the normalized signal $y_{x,\text{norm}}$. The purpose of this normalization was to maintain the original signal amplitude while enabling a fair comparison across the different signals. For each signal and scenario the mean, median, and standard deviation were calculated. The absolute difference between the no-fault and faulty cases for each metric was then computed. Figure 3.1 shows an example of the process of calculating the absolute difference in mean for two signals. This example is meant to illustrate the concept of the method and underscore that a larger absolute difference indicates a greater distinction between no-fault and faulty conditions, facilitating easier detection when the system deviates from its intended behavior. Similar procedures were applied to the other metrics separately for both over- and underfueling cases.

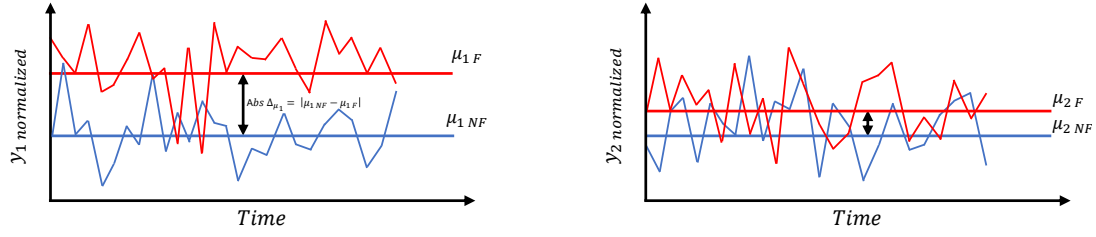


Figure 3.1: Illustration of the absolute difference calculation for the mean, where NF represents the no-fault scenario and F the faulty scenario. In this example, the signal to the left shows greater separation between no-fault and faulty conditions than the signal to the right, which would result in a higher score with this method.

All 23 signals were ranked based on their absolute difference. The signal with the largest absolute difference in each dataset received a rank of one, while the smallest difference received a rank of 23. The ranks for each signal were then summed up across all fault sizes and metrics. This process resulted in a list where the signals with the lowest total rank represented the signals with the most significant combined difference in mean, median, and standard deviation between the fault-free and faulty states across all fault sizes. This was done separately for over- and underfueling. The resulting top ten signals with the largest absolute differences can be seen in Table 3.1. The top-scoring signals are largely similar for each case, with only minor variations observed between them.

Table 3.1: The result for the absolute difference for both over- and underfueling. With the signals in decreasing order with the biggest absolute difference being at the top of the table.

Overfueling	Underfueling
NO _x after catalyst	HPP speed
NO _x before catalyst	HPP current
HPP control signal	HPP control signal
Exhaust brake actuator pressure	NO _x after catalyst
O ₂ before catalyst	Intake manifold temperature
HPP speed	Pressure before turbo
O ₂ after catalyst	O ₂ before catalyst
Wastegate actuator pressure	Average engine speed
P _{exh}	NO _x before catalyst
Intake manifold temperature	O ₂ after catalyst

3.2.2 Feature Importance Using Random Forest

There are many different FI methods available and the FI in RF was used because it combines the advantages of different variable selection techniques and provides insight into which signals contribute the most in an RF classifier. It has also been shown effective in other studies, such as [19] for selecting important features. To understand how FI operates within RF, it is important to first understand the workings of RF itself.

RF is constructed from multiple decision trees that collaborate in making decisions, a process known as ensemble learning [20]. A decision tree operates as a hierarchical structure, navigating through sequential decisions to eventually arrive at a conclusion based on different statements. It consists of a root node, which is the starting point of the decision tree, internal nodes that make feature-based decisions during the splitting process, and leaf nodes where the outcomes or predictions end up. To achieve the purest possible splits, each internal node tries to select the feature that best divides the data. [21]

In Figure 3.2 this process is illustrated in a simplified way. While this only includes one internal node and two leaf nodes, a complete decision tree typically includes a root node and many more internal nodes and leaves. In this example, Signal x is used to split the data, resulting in leaf nodes 1 and 2. As observed, this split yields a purer split when the outcome is false compared to true. The algorithm aims to maximize this purity when choosing splits. In a real and complete decision tree, leaf 1 would likely be further split by another feature to achieve even purer leaf nodes. This process continues until a stopping criterion is reached, such as the maximum depth or the minimum number of samples per leaf. These are hyperparameters that need to be tuned.

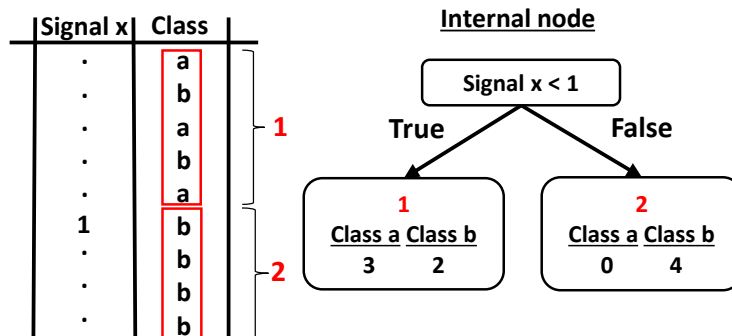


Figure 3.2: A simplified example of how a decision tree works, by showing how an internal node makes a decision based on Signal x and what the outcome would look like in the resulting leaves.

Each tree in the RF is trained on a random subset of the data, a process known as bootstrapping. At each decision split within a tree, only a randomly chosen subset of the features is taken into consideration, a technique known as feature bagging. The model then splits the node with the feature that maximizes the purity. During prediction, data is passed through each tree, and each tree gives a vote on which class it thinks the data point should be in. The final prediction is decided by majority voting from all trees. [22] This results in a predicted probability, and depending on whether the predicted probability is under or above the threshold, the data point is assigned to a specific class.

The construction of the RF during training allows for the assessment of each feature's importance to the model's prediction. This is what is called FI in RF and it is often done by measuring how the total mean decrease in impurity is changed by each feature over all trees in the model. In RF, one way to quantify this impurity is to use the Gini impurity [21]. At each node (τ) in tree (T), the optimal split is based on the resulting Gini Impurity $GI(\tau)$, which measures how effectively a split separates different classes. The Gini impurity is calculated as

$$GI(\tau) = 1 - \sum_{i=1}^K p_i^2 \quad (3.2)$$

where (K) represents the number of classes, and (p_i) represents the proportion of samples in class (i). The resulting impurity decrease (ID) at node (τ) from a split that sends samples to the next nodes (τ_l) and (τ_r), determined by a threshold (J_θ) on the feature (θ), is calculated as (3.3). [23]

$$ID(\tau) = GI(\tau) - p_r GI(\tau_r) - p_l GI(\tau_l) \quad (3.3)$$

The resulting impurity decrease from each split is recorded, and from that, the mean and standard deviation of the accumulation of impurity decrease within each tree can be calculated for each feature. A bigger value of the mean and standard deviation of the impurity decrease indicates that the feature contributes more to reducing the impurity across the forest, and can therefore be seen as more important in making accurate predictions.

The RF classifier was implemented in Python using the `RandomForestClassifier` class from the `scikit-learn` library [24]. Before training the RF model, the fault-free data was labeled with zeros and the faulty data with ones. The data was separated into 80% for training and 20% for testing. It was then normalized using the no-fault training data to obtain a zero mean and a maximum absolute magnitude around one. Both the no-fault and faulty data were concatenated for both the training and test datasets. The over- and underfueling cases were separated to examine how different signals contributed to the detection of the two faults. The hyperparameters were set to their default values, which can be found in the documentation of `scikit-learn`, to achieve a standard RF model. The model was trained through the `fit` function and evaluated using the `score` to assess how well the predictions matched the actual values. This ensured that the classifier achieved a score high enough for the model to be considered valid for evaluating FI. The FI was obtained using the `feature_importances_` function. The results were then plotted to visually identify which features contributed the most to the model's predictions.

In Figure 3.3, the result of computing the FI in the RF can be seen and a larger bar indicates that the signal is more important for the model to make accurate predictions. It was observed that the RF model used for computing the FI achieved above 95% accuracy for both over- and underfueling, which was deemed high enough to consider the model valid.

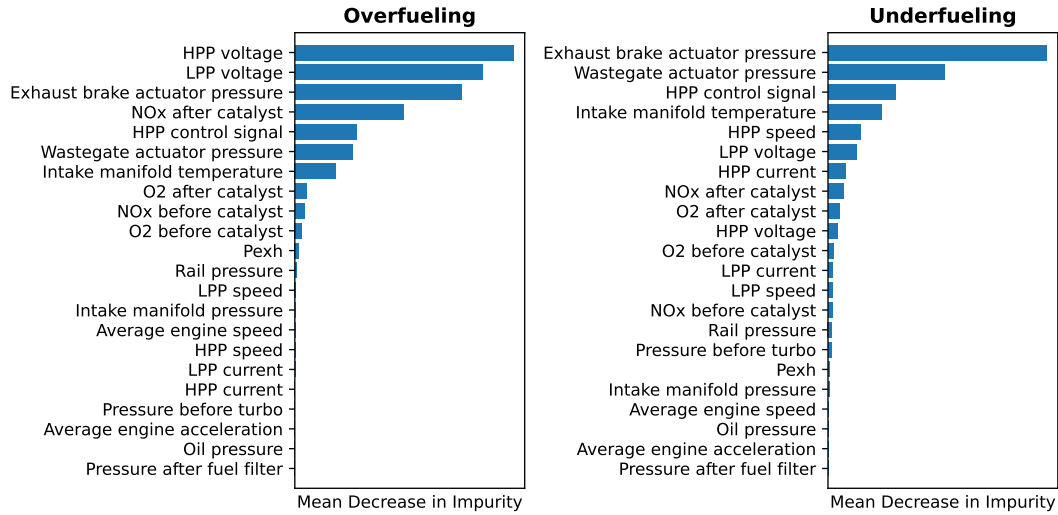


Figure 3.3: The result for the FI for the RF classifier for both over- and underfueling. A larger bar indicates a signal that is more important for the model to make accurate predictions.

3.3 Result of Signal Identification Process

The results from both the absolute difference and FI methods indicate that the signals expected to be important, as discussed in Section 3.1, also achieved high scores in both methods. This alignment suggests that previous research findings are consistent with the experimental data. For instance, exhaust emission signals such as NO_x and O_2 , both before and after the catalyst, scored high in both methods, with slightly higher scores for overfueling compared to underfueling. The same trend was observed for the exhaust pressure signals, which scored high in both methods for overfueling. However, for underfueling, the exhaust brake actuator pressure and wastegate actuator pressure scored high in FI but not in the absolute difference method. The intake manifold temperature signal received high scores in both methods for both over- and underfueling. Additionally, signals connected to the HPP scored high in both methods, with the HPP speed and control signals being particularly significant.

Considering both the literature and insights from the two methods, the NO_x and O_2 signals were selected to proceed with. To minimize the risk of the catalyst affecting the signals, the signals before the catalyst were chosen. Next, the other high-scoring signals were investigated, and it was decided to proceed with one of the exhaust pressure signals. The P_{exh} signal was chosen despite scoring lower than the other two signals. It was deemed more predictable because it is closer to the fault and not near an actuator controlled by other factors, which can cause signal changes and reduce reliability. Furthermore, signals connected to the HPP were not selected, even though the speed and control signals scored high in both methods. This was because the control signal could potentially be influenced by external factors, such as the control strategy, which would in turn affect the speed. This could compromise the signals reliability, making it less suitable for diagnosing faulty injectors. High scores for the intake manifold temperature were also noted, but it was decided not to proceed with this signal. Some tendencies of being affected by how the experiment was performed were observed in the signal, making it less reliable compared to the other selected signals. The outcomes of the signal identification process are summarized in Table 3.2. It can be seen in Figure 2.1 that the selected signals were close to the fuel injectors and also placed after the combustion.

Table 3.2: The target signals selected from the signal identification part.

Signal
NO_x before catalyst
O_2 before catalyst
P_{exh}

When examining the selected target signals, it was observed that the NO_x signal decreased with underfueling and increased with overfueling. This aligned with the findings presented in [18], that less fuel before ignition resulted in a decrease in NO_x emissions while an increase in fuel before ignition resulted in higher NO_x emissions. The behavior of the O_2 signal was observed to be opposite to that of the NO_x signals, which is reasonable given their connection in the combustion processes. While this explanation is simplified and the combustion process in a diesel engine is complex, it highlights the rationale behind these observations. As illustrated in Figure 3.4, it was observed that the exhaust pressure varied in magnitude both during fault and no-fault conditions, exhibiting overall lower pressure with occasional high-pressure peaks. During overfueling, the signal demonstrates prolonged periods of high-pressure peaks compared to the no-fault scenario. Since temperature and pressure are closely correlated, an increase in temperature is likely to lead to an increase in pressure. This aligns with the increased combustion temperature reported in [18] when more fuel was injected before ignition. Meanwhile, the signal appears less affected during underfueling, displaying overall lower values and more instantaneous peaks. These results indicate that the signal is particularly influenced by faults during periods of higher pressure, while it behaves similarly to normal cases during lower pressure intervals. This explains why the exhaust pressure scored lower in the two methods for underfueling.

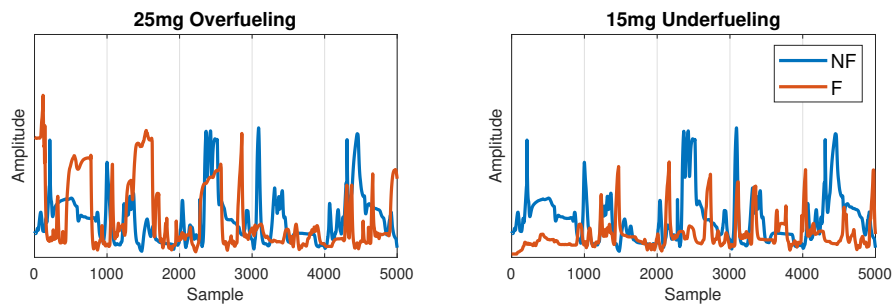


Figure 3.4: Behavior of P_{exh} signals during no-fault, overfueling, and underfueling conditions.

To conclude this chapter, three target signals have been identified as important and informative for detecting both over- and underfueling. These signals will be used in the next chapter as indicators to determine whether an injector is faulty. This will be achieved by predicting how these signals should behave under fault-free conditions and then comparing these predictions with the actual signal.

Chapter 4

Development of Residual Based Diagnosis System

This chapter outlines the necessary theory and methods for developing, implementing, and evaluating the residual-based diagnostic system. It also presents results from the process of selecting inputs for the data-driven residuals.

4.1 Residual Generator

Residuals are commonly used in diagnostic systems to detect faults. A residual r_t can be constructed by comparing the measured output y_t with an estimate of the output $\hat{y}_t$, as illustrated in Figure 4.1. The output estimate can be generated using model-based or data-driven approaches. Under nominal conditions, the residual will be close to zero, as the measured and estimated outputs are in agreement. However, when a fault occurs, the measured output is affected, causing the residual to deviate from zero as illustrated in Figure 4.2. This deviation is what allows the system to detect faults. A residual is sensitive to a fault f_i , if the residual during that fault is $r_{fi} \neq 0$. [25]

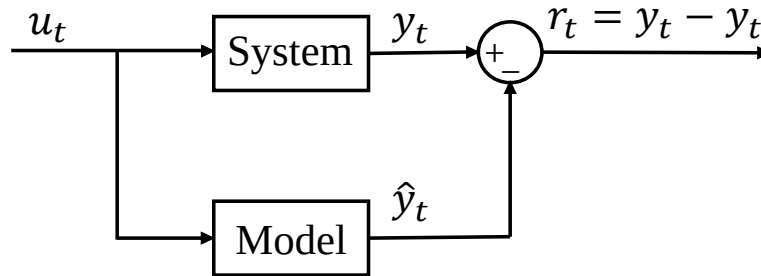


Figure 4.1: A block diagram depicting the equation for constructing the residuals.

4.2 Hypothesis Test

The fault detection problem can be formulated as a hypothesis testing problem. It can be viewed as a binary test, as the hypothesis test will result in one of two possible outcomes: no-fault or fault. The null hypothesis H^0 , represents the default statement implying no fault. The alternative hypothesis H^1 , contradicts the null hypothesis and indicates that there is a fault in the system. To make a decision, a test quantity is constructed. This test quantity $T_t(r)$, is a function derived from the residual which maps to a scalar value that is then compared with

a threshold J . If the test quantity exceeds the threshold, then the null hypothesis is rejected, indicating the presence of a fault. The hypothesis test can be formulated according to

$$H_t = \begin{cases} H_t^0 & \text{if } T_t(r) \leq J \\ H_t^1 & \text{if } T_t(r) > J \end{cases} \quad (4.1)$$

The test quantity may for example be based on a residual generator. [25]

4.3 CUSUM Test Quantity

In this project, the test quantity $T_t(r)$ was created using the CUSUM (CUmulative SUM) algorithm. It is a non-linear detection algorithm that reacts to changes over time, meaning that the test quantity will accumulate and eventually exceed the threshold [25]. This method is great when the residual is noisy, and is a good alternative to a low-pass filter. The CUSUM algorithm is implemented according to

$$T_t(r_t) = \max(0, T_{t-1} + |r_t| - \nu) \quad T_0 = 0 \quad (4.2)$$

where T_t is the test quantity, r_t is the residual, ν is a design parameter, and t is sample index.

The design parameter ν should be greater than the magnitude of $|r_t|$ in the fault-free case, ensuring that $T_t = 0$ for all t . Upon the occurrence of a fault, the term $|r_t| - \nu$ should exceed zero, resulting in a positive accumulation of the test quantity, as illustrated in Figure 4.2, where the dotted line indicates the beginning of a fault. An alarm is generated when T_t exceeds some positive threshold J . However, for this project, a threshold J was not proposed for the CUSUM algorithm.

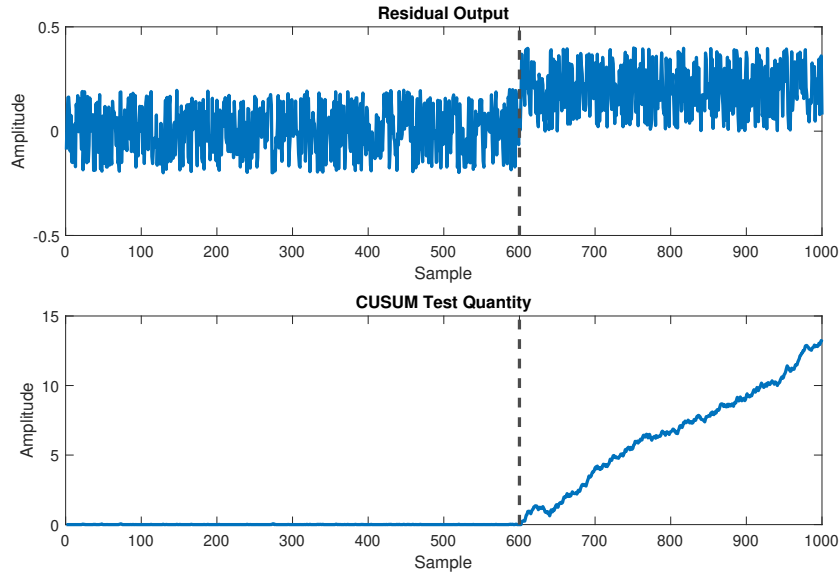


Figure 4.2: Illustrates the behavior of a residual output in a CUSUM algorithm. The dotted line indicates the beginning of a fault.

4.4 Choice of Model Structure

Choosing the appropriate model structure for a data-driven approach can be challenging, as it requires considering both system dynamics and the model's intended purpose. Modeling linear systems is generally less complex and more straightforward compared to modeling nonlinear systems. However, it is essential to take into account the system's physical properties to determine if any critical non-linearities are present. In some cases, non-linear transformations can reveal simpler linear relationships, but if the dynamics are too complex, a non-linear model might be necessary. Sometimes, linearized models can also be effective within bounded operating conditions. [26]

State-Space (SS) models are a mathematical representation of dynamic systems, used to describe their behavior over time. Although SS models are typically associated with linear systems, they can also be used to approximate non-linear systems. In SS models, the system is described by a set of state variables that capture its internal state at any given time. The general structure of a linear SS model is

$$\begin{aligned}\dot{x}_t &= Ax_t + Bu_t \\ \hat{y}_t &= Cx_t + Du_t\end{aligned}\tag{4.3}$$

Here, x_t is the state vector, u_t is the input vector, and $\hat{y}_t$ is the output vector. The matrices A , B , C , and D represent the state matrix, input matrix, output matrix, and feedthrough matrix, respectively.

Autoregressive exogenous (ARX) models are mathematical tools used to describe and predict the behavior of dynamic systems. ARX models rely on a linear relationship with their previous values, meaning they use past output values to predict future outputs. ARX models assume linear relationships between previous outputs, external inputs, and current outputs. This linearity simplifies the modeling process, making it more manageable and computationally efficient. However, despite their effectiveness, ARX models have certain limitations, such as their reliance on linearity, the risk of overfitting, and sensitivity to noise. The general form of the ARX is

$$\hat{y}_t = \sum_{k=1}^n a_k y_{t-k} + \sum_{i=1}^m \sum_{l=0}^n b_{i,l} u_{i,t-l}\tag{4.4}$$

Here $\hat{y}_t$ is the output, $u_{i,t-l}$ is the exogenous input, a_k are the autoregressive coefficients, b_i are the coefficients for the exogenous input, m is the number of signals, and n is the horizon [11].

Non-linear autoregressive exogenous (NARX) models can represent complex non-linear relationships between inputs and outputs. Like ARX models, NARX models incorporate past output values to predict future ones. The complexity of a NARX model can vary depending on the order of the autoregressive terms, the order of exogenous inputs, and the type of non-linear functions used. The mathematical representation of the NARX model is

$$\hat{y}_t = g(y_{t-1}, \dots, y_{t-n}, u_t, \dots, u_{t-m})\tag{4.5}$$

where g is the function, y is the output, u is the input, m is the horizon for the output, and n is the horizon for the input. [26]. The inputs to the function g are called regressors, and they can be either linear or non-linear.

4.5 Construction of Data-Driven Residuals

The residuals were formulated as previously described in Figure 4.1, where the output $y(t)$ represents a sensor output. Hence, the residuals are calculated as the predicted signal subtracted from the actual signal. The total number of residuals depends on the number of identified signals mentioned in Chapter 3, with one residual created for each identified target signal. To construct the prediction model, a data-driven approach was used as opposed to a model-based approach.

The prediction was derived through the System Identification Toolbox in MATLAB [27], where the system dynamics are estimated using the SS, ARX, and NARX models. For each target signal NO_x , O_2 and P_{exh} three predictions were obtained to evaluate which model is most suitable for diagnostics. The dataset utilized to generate these predictions consisted of fault-free data without a load and was divided into 70% training and 30% validation. The data was normalized to have zero mean and an appropriate maximum absolute magnitude around one. To obtain the predictions, the training data was utilized along with the sample time.

The SS model was constructed using the function `ssest`, which estimates a continuous-time state-space model with an order denoted by nx , representing the number of states. As the model order was unknown, it was determined by analyzing the relative measure of how much each state contributes to the input-output behavior of the model (log of singular values of the covariance matrix), as illustrated in Figure 4.3. A higher value of *Log of Singular values* indicates a greater contribution. The number of nx should be chosen such that the state provides the most significant contribution, where the red bar represents the optimal recommendation according to the *Model Order Selection* in the toolbox. A significant decrease in a state's contribution suggests that the number of nx preceding the drop is sufficient to obtain an accurate model. The model orders 1 to 10 were evaluated, where the optimal recommendation was applied for each target signal.

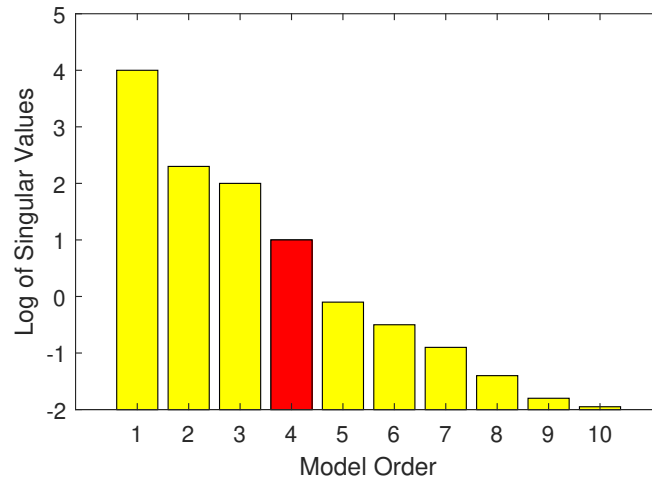


Figure 4.3: Figure illustrating the *Model Order Selection*, where the vertical axis indicates the contribution to the input-output behavior of the model for each state, and the horizontal axis represents the model order. The red bar indicates the optimal recommendation.

The ARX model was constructed using the function `arx`, which estimates the coefficients of A and B polynomials along with their standard deviations. The orders specified by `ORDERS = [na nb nk]` determine the dimensions of the A and B polynomials in the ARX model. To establish these orders, a range of values from 1 to 5 was set, each with corresponding dimensions. Then they were evaluated based on the model's NRMSE performance.

Lastly, the NARX model was created using the function `nlarx`, which estimates a model that utilizes the output function FCN to map the regressors to the model output. For this project, a combination of linear weights and an offset was used to compute its output. The vector of linear regressors R_{Linear} , inputted to the model comprised the two previous values of the output, along with the current value and the two previous values of the inputs, according to

$$R_{Linear} = \begin{pmatrix} y_{t-1} \\ y_{t-2} \\ u_{1,t-1} \\ u_{1,t-2} \\ u_{1,t-3} \\ \vdots \\ u_{i,t-3} \end{pmatrix} \quad (4.6)$$

where y is the output, u is the input, t is the sample index, and i represents the number of inputs for each target signal. Alongside the linear regressors, polynomial regressors were also utilized, with their selection based on experiments due to limited system knowledge.

To validate the model, the System Identification Toolbox's `compare` function was utilized, comparing the system model response with the measured validation data, as illustrated in Figure 4.4. The normalized root mean square error (NRMSE) was used as a measure of the quality of fit. The model's accuracy is displayed in the legend as a percentage, with 100% corresponding to a perfect prediction. A deeper explanation of how the model's accuracy is defined can be found in the documentation for the System Identification Toolbox [27].

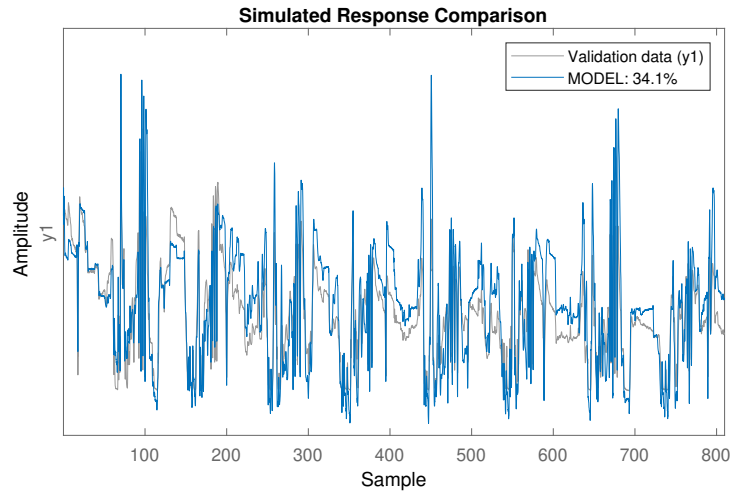


Figure 4.4: An example of model validation using the `compare` function, which compares model responses with measured data. Showing the NRMSE as a measure of the quality of fit in the legend.

4.6 Input Selection for Each Target Signal

When predicting the target signals, the inputs to feed the model had to be determined. The decision was based on previous literature, existing system knowledge, and a feature importance algorithm, which is explained in Section 3.2.2.

When predicting the NO_x signal in a diesel engine, the signals presented in Figure 4.5 were used, as described in the paper [28]. The signals available in the dataset for this project included the engine speed n_e , injected fuel mass flow $\dot{m}_f$, rail pressure p_{rail} , and the intake manifold absolute pressure which is assumed to be equivalent to the intake pressure p_{im} . However, the remaining signals which include the main injection fuel quantity mi_{QNT} , the pilot injection fuel quantity pi_{QNT} , main start of injection mi_{SOI} , and the pilot start of injection pi_{SOI} were not available in the dataset and were therefore not utilized for predicting NO_x in this project.

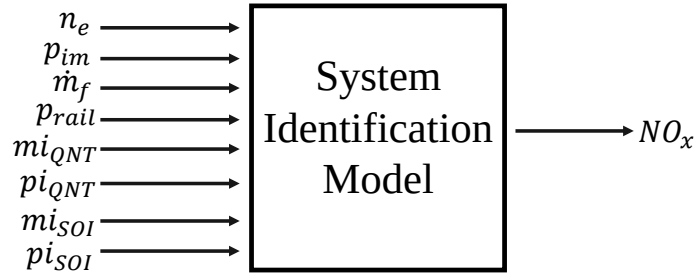


Figure 4.5: Inputs used to predict the NO_x signal, as described in [28].

The prediction of the O_2 signal was influenced by the previous research [13], which suggested that the reconstructed injection mass $m_{f,rec}$ can be calculated using the measured oxygen concentration v_{O_2} , and the air mass m_{air} according to

$$m_{f,rec} = \frac{m_{air}(1 - 4.8v_{\text{O}_2})}{14.5 + 4.6v_{\text{O}_2}} \quad (4.7)$$

In the article, the reconstructed injection mass refers to the injection mass calculated from the measured oxygen concentration, hence assuming that $m_{f,rec}$ is equivalent to m_f . Additionally, the concentration v_{O_2} is assumed to be the measured O_2 signal. Under the assumption that the intake manifold and the cylinders have the same pressure and temperature, m_{air} can be calculated according to

$$m_{air} = \frac{p_{im}V_d}{R_{air}T_{im}} \quad (4.8)$$

where p_{im} is the intake manifold pressure, V_d is the displaced volume, R_{air} is the specific gas constant for air and, T_{im} is the intake manifold temperature [29]. With these equations, it is possible to get an estimate of which signals the oxygen is dependent on.

Predicting the P_{exh} was influenced by the book [30], where the dynamic exhaust pressure p_{em} , in a combustion engine is calculated according to

$$\frac{dp_{em}}{dt} = \frac{R_{exh}T_{em}}{V_{em}}(\dot{m}_{exh} - \dot{m}_{cat}) \quad (4.9)$$

where R_{exh} is the specific gas constant for exhaust gas, T_{em} is the exhaust manifold temperature, and V_{em} is the exhaust manifold volume. Here the exhaust mass flow $\dot{m}_{exh}$, catalyst mass flow $\dot{m}_{cat}$ and air mass flow $\dot{m}_{air}$ are computed according to

$$\dot{m}_{exh} = \dot{m}_f + \dot{m}_{ac} \quad (4.10)$$

$$\dot{m}_{cat} = \sqrt{\frac{p_{es}}{C_0 R_{exh} T_{es}}} (p_{es} - p_{amb}) \quad (4.11)$$

$$\dot{m}_{air} = \eta_{vol} \frac{p_{im} V_d n_{cyl} n_e}{R_{air} T_{im} n_r} \quad (4.12)$$

respectively. Where $\dot{m}_{ac}$ is the cylinder air mass flow, p_{amb} is the ambient pressure, p_{es} and T_{es} is the exhaust system pressure and temperature respectively, n_{cyl} is the number of cylinders, n_r is the number of revolutions per cycle and, η_{vol} is the volumetric efficiency. For this project, it is assumed that the exhaust pressure will exhibit similar behavior in a diesel engine. Furthermore, an examination of the feature importance reveals that the oxygen concentration before catalyst O_2 , and the exhaust brake actuator pressure p_{eba} rank highly.

The final results of the input selection for each target signal NO_x , O_2 , and P_{exh} , is presented in Table 4.1. These input signals were chosen based on their correlation with the respective target signals, as indicated by the equations above. The NO_x was predicted using the signals m_f , n_e , p_{rail} , and p_{im} . O_2 was predicted using m_f , T_{im} , and p_{im} . While P_{exh} was predicted using m_f , n_e , O_2 , and p_{eba} .

Table 4.1: The input signals used to calculate the NO_x , O_2 , and P_{exh} signals.

Input Name		
NO_x	O_2	P_{exh}
Injected fuel mass, m_f	Injected fuel mass, m_f	Injected fuel mass, m_f
Average engine speed, n_e	Intake temperature, T_{im}	Average engine speed, n_e
Rail pressure, p_{rail}	Intake pressure, p_{im}	Oxygen before catalyst, O_2
Intake pressure, p_{im}		Exh. brake actuator pressure, p_{eba}

Chapter 5

Development of Random Forest Classifier

This chapter outlines the method for developing and evaluating an ML-based diagnostic system. This work will focus on the ML method called RF classifier, which has demonstrated promising results in diagnosing engine faults in previous research [31]. RF has also shown effectiveness in fault classification for diesel engines [32], further supporting its suitability for the work. The necessary theory for understanding RF is presented in Section 3.2.2.

5.1 Data Preparation

Table 5.1 shows the input signals used in the classifier, including those from the data-driven residuals, as well as the NO_x , O_2 , and P_{exh} signals. These signals were used so that the classifier received similar information as the data-driven residuals to ensure a fair comparison.

Table 5.1: Input signals used in the RF classifier.

Input Name
Average engine speed, n_e
Exhaust brake actuator pressure, p_{eba}
Exhaust Pressure, p_{exh}
Intake pressure, p_{im}
Intake temperature, T_{im}
NO_x before catalyst, NO_x
Oxygen before catalyst, O_2
Rail pressure, p_{rail}
Injected fuel mass, m_f

The data was separated into 80% for training and 20% for testing. It was then normalized based on the no-fault training data, to have a zero mean and an appropriate maximum absolute magnitude around one. The over- and underfueling cases were separated into two different fault classes. The next step involved preparing the data for the different ways it would be fed to the classifier. This was done through four different approaches, which were implemented independently of each other. The following approaches were considered:

- **Raw Input:** The classifier received raw measurements, classifying one sample from each signal at a time.
- **Batch Input:** The classifier was fed with batches of data, which included measurements from previous time steps.
- **Statistical Input:** The classifier was provided with statistical measurements, which included the mean and standard deviation from segments of data.

- **Residuals as Input:** The classifier was combined with the data-driven residuals, by feeding the three residuals into the classifier.

For the Batch Input, the data were prepared as illustrated in Figure 5.1. In this representation, f corresponds to signal 1, and g represents signal x , which denotes the last signal in each batch. The variable t indicates the time instance, while m represents the last time step used in each batch. Each row in the figure represents a batch. Using this approach yielded some batches (m no. of batches) containing empty values. To address this, the rows containing these empty values were removed. It was determined to incorporate 50 previously recorded measurements in each batch, aiming to capture a relevant time window without overly increasing the batch size. With 9 measurement signals, each batch contained 450 samples. This method allowed for the inclusion of 5 seconds of the system's behavior into each decision-making process.

$$\begin{array}{c}
 \text{Signal 1} \qquad \qquad \text{Signal X} \\
 \text{Batch 1} \left\{ \begin{array}{c} \overbrace{f_{1,t}, f_{1,t-1}, \dots, f_{1,t-m}} \\ \overbrace{g_{1,t}, g_{1,t-1}, g_{1,t-m}} \end{array} \right. \\
 \text{Batch 2} \left\{ \begin{array}{c} \overbrace{f_{2,t}, f_{2,t-1}, \dots, f_{2,t-m}} \\ \overbrace{g_{2,t}, g_{2,t-1}, g_{2,t-m}} \end{array} \right. \\
 \vdots \\
 \text{Batch N} \left\{ \begin{array}{c} \overbrace{f_{N,t}, f_{N,t-1}, \dots, f_{N,t-m}} \\ \overbrace{g_{N,t}, g_{N,t-1}, g_{N,t-m}} \end{array} \right.
 \end{array}$$

Figure 5.1: The data preparation for batches containing measurements from the previous time steps.

When the model was fed with statistical input, the data was first trimmed to ensure the row count was divisible by the segment size. Subsequently, the data was divided into segments, and the mean and standard deviation were computed for each segment. These calculated values were then concatenated for each segment. This process is illustrated in Figure 5.2. The segment size was set to 300 samples, representing measurements acquired within a 30-second time frame. This decision aimed to find a balance between accumulating measurements over time and avoiding too long duration that might lead to misses in important changes in the data.

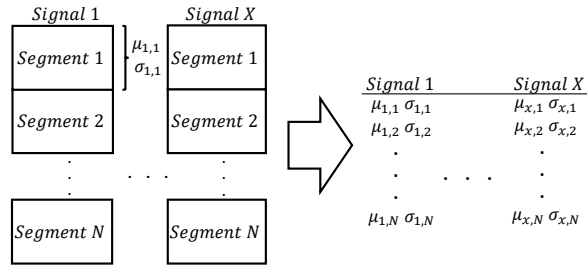


Figure 5.2: The data preparation for when the mean and standard deviation of segments were calculated.

The RF classifier fed with data-driven residuals was aimed to be an alternative method of utilizing information from the previously used signals since the three residuals include information from all those signals. When feeding the residuals into the model, they were treated

as raw measurements to ensure a fair comparison with the original data. Tests by preprocessing the residuals like the measurements were also conducted and those results will only be presented in text.

5.2 Implementation

The implementation of the RF classifier was done in Python using the `RandomForestClassifier` class from the scikit-learn library. A new model was trained for each specified method of feeding data into the classifier. As mentioned earlier, it was decided to separate the over- and underfueling cases to gain a better understanding of how the classifier performed across the two different scenarios. The hyperparameters of the RF classifier were optimized using Grid Search, implemented with the `GridSearchCV` function from the scikit-learn library. The hyperparameters were optimized for each specified new model. The following hyperparameters were considered in the grid search:

- **n_estimators:** The number of trees in the model, varied between 50, 100, 150, and 200.
- **max_depth:** The maximum depth to which each tree could extend, varied between None (no maximum depth), 10, 20, and 30.
- **min_samples_split:** The minimum amount of samples necessary to split a node, varied between 2, 4, 6, and 8.
- **min_samples_leaf:** The minimum amount of samples necessary at a leaf node, varied between 1, 2, 4, and 8.

The `GridSearchCV` function was initialized with the RF classifier as the estimator, the defined parameter grid, and a 5-fold cross-validation. It then performed cross-validation for each combination of hyperparameters, trained the model, and evaluated it. The best set of hyperparameters, those that yielded the highest cross-validation score, were used to build the RF model.

5.3 Evaluation

To evaluate the model's performance, the RF classifier was fed with the test data. This allowed for the calculation of the overall accuracy, which measures how well the model's predictions match the actual values. It was calculated as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5.1)$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives and FN is the number of false negatives. [33]

Due to some data imbalance, the F1 score was also utilized. It uses both the precision and recall scores of a model. The F1-score is calculated as the harmonic mean of precision and recall as. [34]

$$F1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (5.2)$$

When evaluating the classifier's performance across different operating points, the previously trained models were fed the data representing the new operating conditions. This data included a no-fault scenario under load conditions and two faulty scenarios of 10 mg over- and underfueling under load conditions during aggressive driving. The objective was to investigate the model's ability to generalize on unseen data from other operating conditions. This evaluation was performed on all four methods by feeding the models the test data

representing the different operating conditions. To evaluate the model's performance when exposed to the data representing the different operating conditions, the previously trained models were retrained on the training data, now including this new data, and then evaluated again on the test data representing the different operating conditions.

5.4 Data Visualization

To visualize the classifier's performance over a time series of data, the predicted probabilities were plotted against the true class. This visualization made it easier to understand the model's performance in real-time situations and identify instances of false alarms and missed detections.

Visualizing high-dimensional data can be challenging, especially when working with nine measurement signals. To address this, the unsupervised non-linear technique called t-SNE (t-distributed Stochastic Neighbor Embedding) was used. This method can reduce the dimensionality of high-dimensional data while preserving local structures by utilizing the Kullback-Leibler divergence as a similarity metric, which makes samples that exhibit similarity in the original space remain closely grouped. It is important to note that the dimensionality reduction is achieved through nonlinear optimization, resulting in slightly different results with each run. [32]

By using the `TSNE` function from the *sklearn.manifold* module in Python, a two-dimensional representation of the data suitable for visualization was obtained. These plots provided insights into the separability of the data, making it possible to see how well the performance in detecting faulty behavior should be. For instance, if the data overlaps in the lower-dimensional space, it indicates overlapping in the original space, making it challenging to distinguish between no-fault and faulty behavior.

Chapter 6

Results

This chapter presents the results of the evaluation of two diagnostic methods: Data-driven Residuals and Classification. It also provides the results of the operating point assessment.

6.1 Data-Driven Residuals

A total of three residuals were created, one for each identified signal: NO_x before the catalyst, O_2 before the catalyst, and exhaust pressure. The results of the diagnostic performance for faulty injectors for each residual are presented in this section. Further investigation of the data-driven residuals showed that the different models used to create them exhibited similar behavior and results, including prediction during fault data, residual appearance, and ability to detect faults. Therefore, only results from the SS model will be presented in this section, as they represent the overall results for all models unless otherwise is stated.

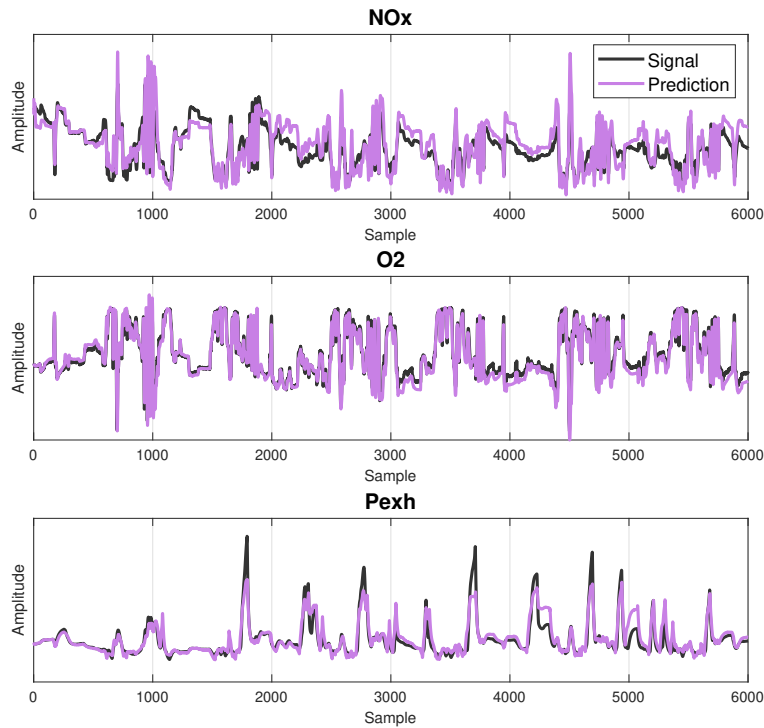


Figure 6.1: The validation of the SS-model's predictive capabilities for the NO_x , O_2 , and P_{exh} signals.

The signal predictions were calculated employing the SS, ARX, and NARX models, as described in Section 4.1. Figure 6.1 shows the validation of the SS model, showcasing its ability to capture the overall system dynamics across all three signals. Although a diesel engine has non-linear characteristics, the linear SS model proves to have a good prediction of the signals. The prediction of exhaust pressure relied on a filtered signal already available in the dataset.

Table 6.1 shows the predictive performance of the models, evaluated using the NRMSE as described in Section 4.5. When comparing the models it was evident that the NO_x prediction did not perform as well, achieving an accuracy of only around 35%, while the O_2 prediction achieved the highest. However, despite the lower accuracy, the NO_x model still effectively captured the dynamics. For both NO_x and P_{exh} , the NARX model provided the most accurate predictions, which is expected considering the system's inherent non-linearities. Whereas, the O_2 signal gave a more unexpected outcome and exhibited the best predictions using the SS model. Despite varying predictive capabilities ranging from approximately 35% to 75%, all models effectively captured the underlying dynamics.

Table 6.1: The predictive accuracy of the SS, ARX, and NARX models for the NO_x , O_2 , and P_{exh} signals, calculated with NRMSE as the measure of the quality of fit.

Model Validation			
	NO_x	O_2	P_{exh}
SS	34.7 %	74.8 %	58.9 %
ARX	33.0 %	59.1 %	45.5 %
NARX	48.8 %	72.1 %	74.3 %

When the models are provided with faulty data, it is expected that the prediction will deviate from the actual signal. The extent of the deviation can indicate the effectiveness of fault detection, with a larger deviation being desired. This deviation is observed for the signals NO_x and O_2 across all three models. Additionally, it is noted that the SS, ARX, and NARX models demonstrate similar behaviors in predicting these signals during a fault, with differences primarily seen in the extent of the deviations. During overfueling, the predicted NO_x value increases compared to the actual signal, whereas it decreases during underfueling. Conversely, the O_2 prediction shows the opposite trend. However, for the exhaust pressure, the results vary among the models. Interestingly, the ARX model appears to be less affected by faults and maintains good prediction accuracy of the signal despite system faults. This finding suggests that the ARX model is unable to detect faults using the P_{exh} signal, a conclusion later confirmed by the CUSUM test. In contrast to the ARX model, both the SS and NARX models exhibited deviations from the actual signal. However, these models achieved more accurate predictions in certain intervals despite the presence of faults. This observation may be attributed to the signal behavior, as discussed in Section 3.3, where the signal exhibited similar patterns in certain intervals for both no-fault and fault cases.

In Figure 6.2, the SS-based residuals for 25mg overfueling and 15mg underfueling are presented alongside the no-fault residual, where r_x is the residual and x represents the target signal. These two fault scenarios represent the behavior of the remaining fault sizes within their respective fault classes. As observed, the residuals during no-fault are centered around zero, with some outliers. These deviations represent instances where the model might struggle to capture certain system dynamics. Notably, the exhaust pressure residual tends to exhibit a higher rate of deviations.

Overall, it is observed that the residuals exhibit much larger deviations from zero during overfueling compared to underfueling. However, some smaller separation is also observed during underfueling. Additionally, the NO_x and O_2 residuals display contrasting behaviors between the two fault classes. Analyzing the P_{exh} residual reveals distinct behavior between overfueling and underfueling. Notably, the residual is significantly more affected by overfueling, where the residual exhibits prolonged peaks of larger magnitude. During underfueling, the peaks are much smaller and tend to alternate between positive and negative values. Additionally, the mean value during underfueling is smaller compared to no-fault. Furthermore, in both fault cases, the residual tends to approach values around zero at certain time points, resembling no-fault values.

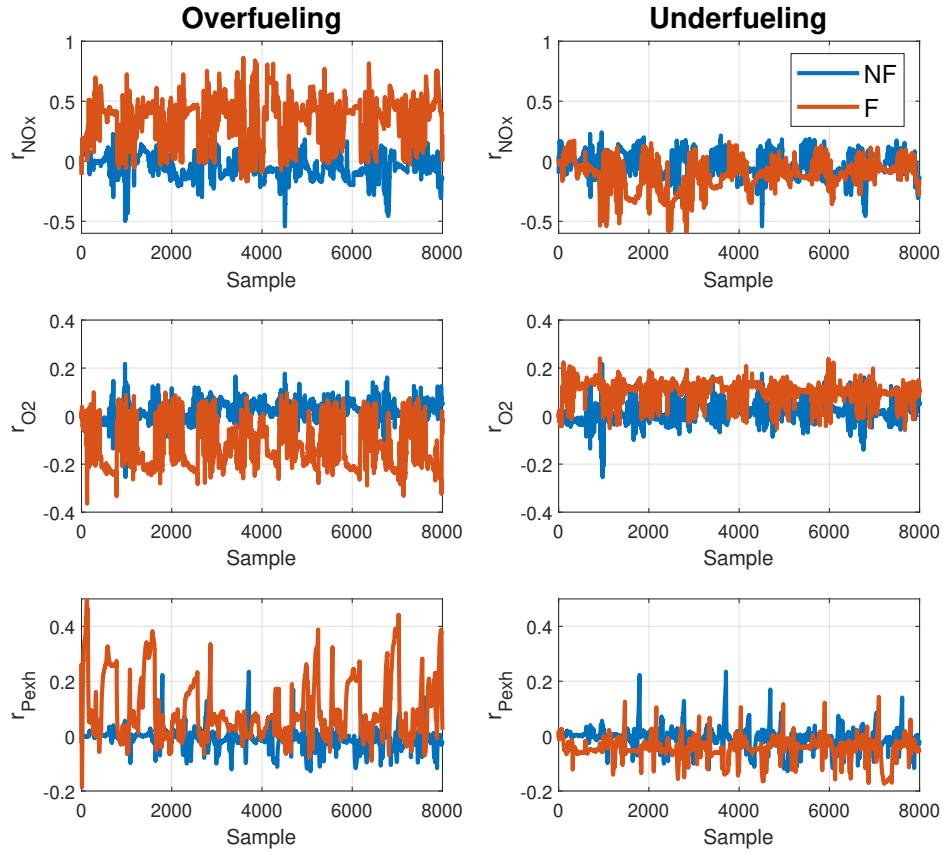


Figure 6.2: The SS-based residuals for no-fault, overfueling, and underfueling for the NO_x , O_2 , and P_{exh} signals.

Histograms of the SS-based residuals are presented in Figure 6.3, showing the distribution for all fault sizes, where r_x is the residual and x represents the target signal that is predicted in the residual. When analyzing the results, it becomes evident that a larger separation between no-fault and fault is observed for larger fault sizes, indicating that detection performance correlates with fault size. Consistent with previous findings, a significant separation is evident between no-fault and overfueling, suggesting that the target signals can effectively be used for fault detection. However, distinguishing between no-fault and underfueling proves more challenging, particularly for NO_x , where a complete overlap exists between 10mg and no-fault scenarios. Despite the clear overlap between no-fault and fault scenarios in both cases, the presence of even a small separation suggests that it is possible to detect faults using the CUSUM algorithm.

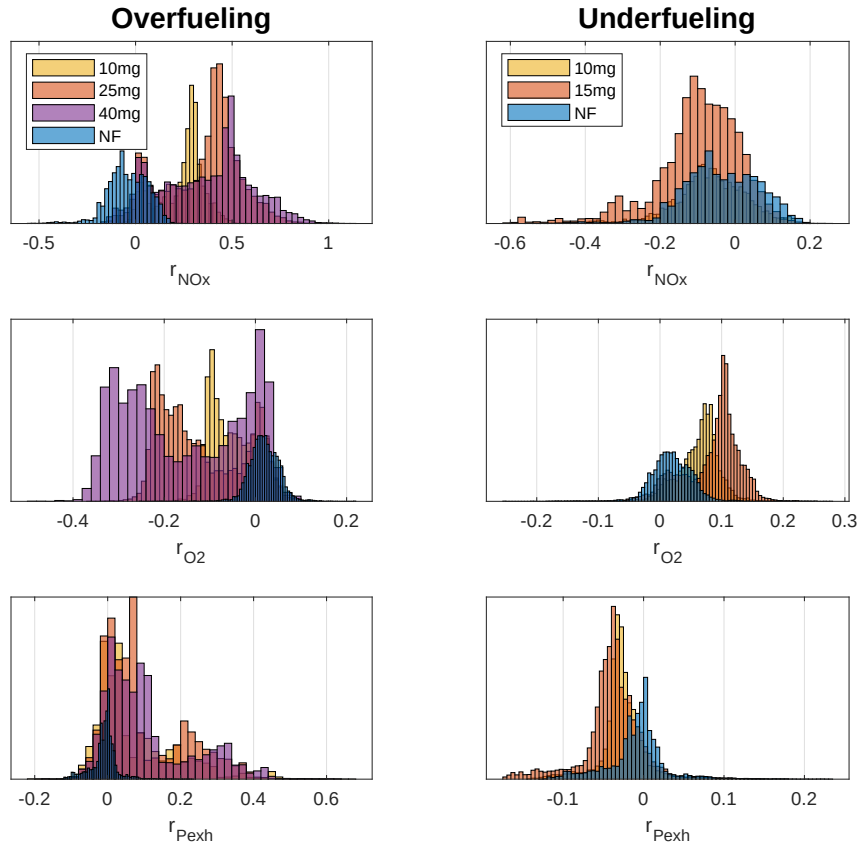


Figure 6.3: Histograms showing the distribution of the SS-based residual for the fault sizes in overfueling, underfueling, and no-fault conditions, analyzed for the NO_x , O_2 , and P_{exh} signals.

The CUSUM test generated using the SS-based residuals is presented in Figure 6.4, where $T(r_x)$ is the test quantity of the corresponding residual from each target signal, x . The CUSUM test yielded expected outcomes based on the behavior of both the residuals and the raw signal during faults. As observed, both NO_x and P_{exh} can effectively detect overfueling regardless of fault size. They exhibit an inability to detect underfueling, as evidenced by the lack of response to the 10mg fault size. During the 15mg fault size, the NO_x test quantity experiences a slight increase before returning to zero, while the pressure test quantity reaches values around 80, suggesting potential sensitivity to larger faults. The O_2 residual can detect both overfueling and underfueling. As observed, the O_2 test quantity for 10mg overfueling appears lower than for other faults. However, the maximum value reaches 100 during a fault compared to only 5 during no-fault conditions, indicating that the fault is detectable.

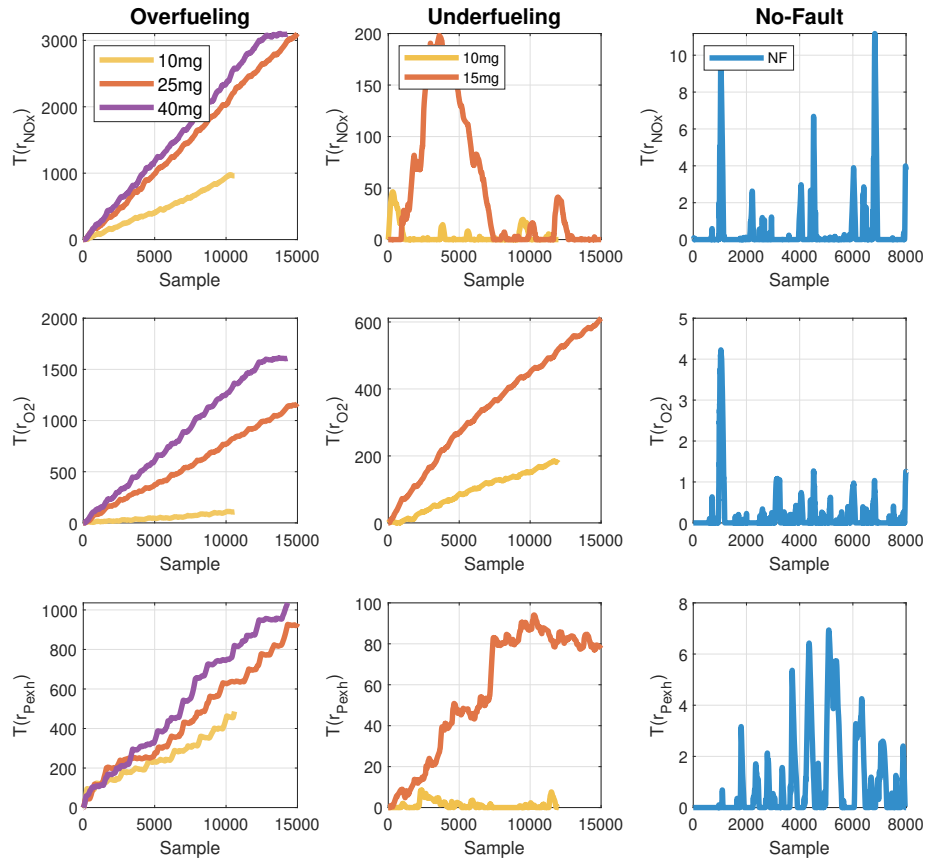


Figure 6.4: CUSUM test illustrating the test quantity derived from the SS-based residual for different fault sizes in overfueling, underfueling, and no-fault conditions, analyzed for the NO_x , O_2 , and P_{exh} signals.

6.2 Random Forest Classifier

In this section, the results obtained from the RF classifiers using different inputs are presented. The resulting predicted probabilities for overfueling for the different inputs are shown first, followed by the predicted probability for the underfueling case. Finally, the resulting t-SNE plots will be shown for both over- and underfueling.

The statistical input naturally contains fewer samples compared to the other methods. This is an important factor to consider when comparing the performance measurements of the statistical input to the other methods. Because of the smaller sample size, false alarms and missed detections will have a greater impact on the overall performance, affecting the accuracy and F1-score more significantly. The comparison will therefore primarily be done by inspecting the predicted probabilities. The accuracy and F1-score will serve more as measures of the overall performance for each input method. If there is an imbalance in no-fault and faulty samples, it's wise to be cautious of the accuracy metric. In such cases, the F1-score can provide a more balanced measure of the model's performance, as it uses both precision and recall.

In the predicted probability plots, each blue dot represents the resulting probability for a sample, indicating the classifier's confidence in classifying them as faulty. The red line represents the threshold, set to the default value of 0.5. If the classifier's probability for predicting a faulty condition, class 1, is above this threshold, the sample will be labeled as faulty. For samples without faults, the blue dot should be below the threshold, while for faulty behavior, it should be above it. A blue dot above the threshold for no-fault results in a false alarm, and one below it for the faulty case leads to a missed detection. Adjusting the threshold allows tuning the ratio of missed detections to false alarms, depending on which is more costly. The threshold in this case was kept at the default value of 0.5 to give equal weight to false alarms and missed detections.

6.2.1 Overfueling

The predicted probabilities for the different inputs during overfueling are shown in Figure 6.5. The plots are divided by fault sizes, marked by vertical black dashed lines. Moving from left to right, the sections represent no-fault, 10mg, 25mg, and 40mg overfueling. Starting from the top of the figure, the use of raw inputs showed promising results. However, it yields both false alarms and missed detections, indicating uncertainty in the model and it struggles to classify the correct class. The performance across the different fault sizes is relatively consistent, although there is some uncertainty towards the end of the 40 mg case.

The batch input made a significant improvement on the performance of the classifier. False alarms were reduced to zero, and missed detections were minimized, showing that this method performs well across the different fault sizes with a bit of struggle at the end of the 40 mg case. This result shows that integrating previous measurements can improve the classifier's performance without adding more measurement signals. However, when doing this, both training and prediction times increased a significant amount.

For the statistical input, the classifier performed well, with zero false alarms and zero missed detections. The predicted probability was low for the no-fault case and high for the faulty case, indicating a confident model. Similar to the other inputs, some uncertainty was observed at the end of the no-fault and 40 mg cases. When using this type of input, both the training and prediction time became faster compared to the other input methods.

The results of feeding the classifier with the three residuals as inputs can also be seen in

Figure 6.5. This alternative method of providing information from the system to the classifier did not improve the performance compared to using the measurement signals directly. However, when the residuals were processed in the same manner as the measurement signals, it yielded promising results, leading to a significant performance improvement. Furthermore, the plot reveals an uneven distribution of no-fault and faulty data for the residuals compared to the measurement signals. This uneven distribution arises from the training process of the models used to generate the data-driven residuals. The models were trained only on the no-fault data, leaving a smaller amount available for predicting the no-fault case and creating the no-fault residual.

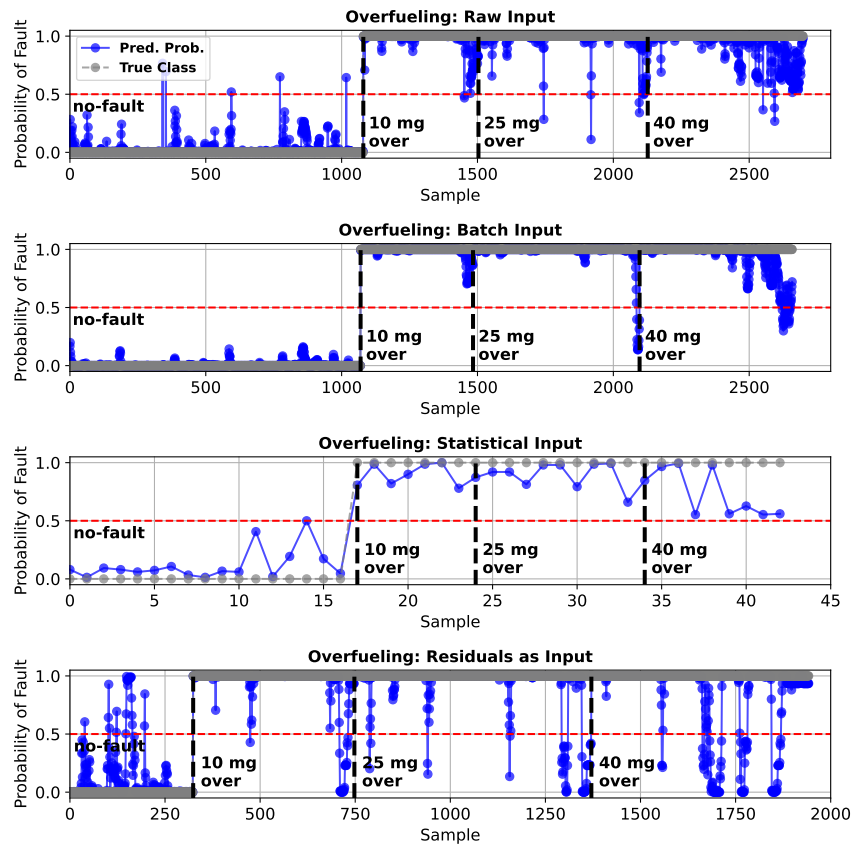


Figure 6.5: The predicted probability for the different inputs to the classifier during overfueling.

In Table 6.2, the performance for the different input methods is summarized. The classifier's performance in detecting overfueling showed promising results, some false alarms and missed detections were observed. The high accuracy and F1-score suggest relatively few misclassifications compared to correct classifications. As seen in Table 6.2, feeding the model with residuals instead of measurements led to worse performance, despite containing similar information of the system. The uneven distribution of data may explain why the models struggle more with the no-fault data compared to the faulty data. When examining the F1-score, particularly suitable for unevenly distributed data, it can be seen that the score is lower for the no-fault compared to the faulty case.

Table 6.2: F1-score and Accuracy for the different input methods during overfueling.

Input:	Raw	Batch	Statistical	Residual
F1-score for No-fault (0.0):	0.99	0.99	1.00	0.76
F1-score for Fault (1.0):	0.99	0.99	1.00	0.94
Weighted Avg:	0.99	0.99	1.00	0.90
Accuracy:	0.9893	0.9904	1.00	0.9043

6.2.2 Underfueling

In Figure 6.6, the predicted probability for different input methods to the classifier is presented during underfueling. Similar to the overfueling case, the plots are segmented by fault sizes, denoted by vertical black lines. Moving from left to right, the sections represent no-fault, 10mg, and 15mg underfueling. Here, class 1 represents a faulty state due to underfueling instead of overfueling. The raw input for the underfueling case shows good results in detecting faulty behavior for all fault sizes. However, the classifier generates slightly more false alarms in the no-fault case compared to the overfueling case.

The performance improved when the classifier was fed with the batch input instead of the raw input. This reduced false alarms and minimized missed detections to zero. Similar to the overfueling case, both the training and prediction times significantly increased with the batch input. When the classifier was fed with the statistical input, the model performed slightly worse than with the batch input. The predicted probabilities hovered around the threshold for the no-fault state, resulting in more false alarms. For the faulty state, the model performed better with few missed detections. The model performed slightly better for 10mg underfueling compared to 15mg.

In Figure 6.6, similar to the overfueling case, it can be seen that using residuals as inputs yields results almost as good as feeding the raw input. Furthermore, it can be seen that there are about the same number of false alarms as missed detections, and the model performs consistently across different fault sizes. Additionally, processing the residuals similarly to the measurement signals showed good results. Similar to the overfueling case, there was an uneven distribution of data for no-fault and faulty states.

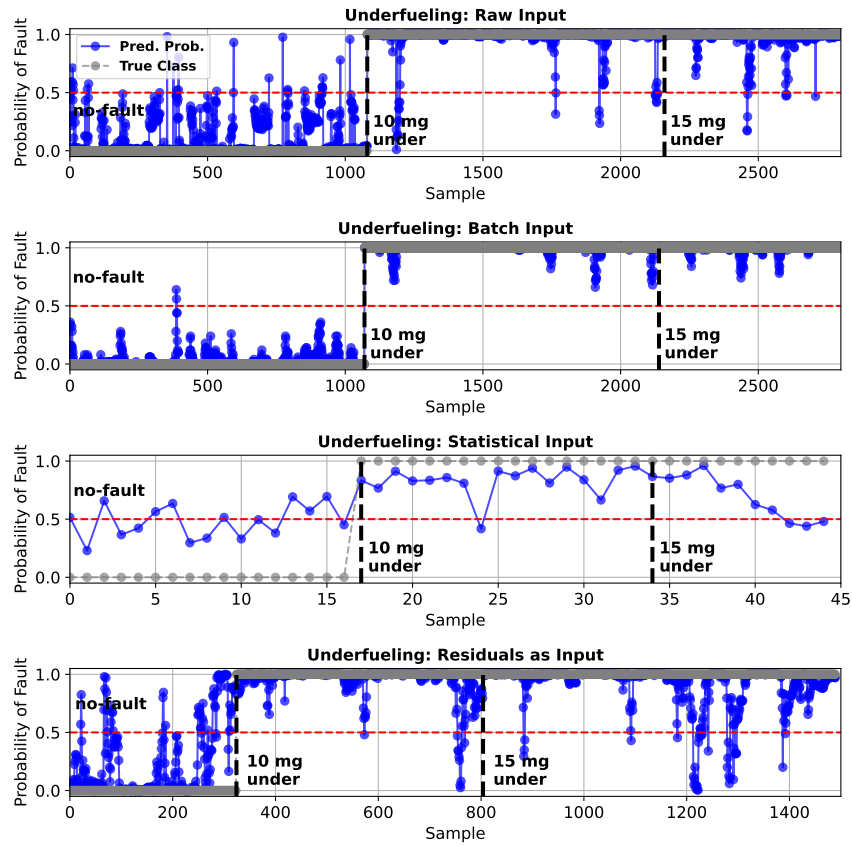


Figure 6.6: The predicted probability for the different inputs to the classifier during underfueling. The red line indicates the threshold.

In Table 6.3, the performance for the different inputs is summarized. The raw input and batch input shows similar performance for both under- and overfueling. The statistical input demonstrates significantly lower performance compared to the other methods. As mentioned previously, this is because fewer samples result in misclassifications carrying greater weight in the overall performance. Upon examining the predicted probabilities, it becomes evident that the performance of this method is not as poor as indicated solely by the performance metrics. It can be observed in the table that using residuals as inputs for the underfueling case yields a higher score compared to when using them for the overfueling case.

Table 6.3: F1-score and Accuracy for the different input methods during underfueling.

Input:	Raw	Batch	Statistical	Residual
F1-score for No-fault (0.0):	0.97	0.99	0.60	0.81
F1-score for Fault (1.0):	0.98	0.99	0.80	0.95
Weighted Avg:	0.98	0.99	0.70	0.92
Accuracy:	0.9806	0.9991	0.7333	0.9220

6.2.3 Data Visualization

The t-SNE plots were used to visualize the relationship between the no-fault and faulty data in the original high-dimensional space. The x and y-axes in the t-SNE plot represent the transformed dimensions resulting from t-SNE's nonlinear dimensionality reduction technique. Figure 6.7 shows the resulting t-SNE plots, illustrating how the no-fault and faulty data are distributed and separated from each other, for the different methods. In the plot, it can be seen that the separation becomes more noticeable as the data is processed, although to a lesser extent for the underfueling case. This trend aligns with the results from the classifier, which demonstrated improvements when the data was processed. There are instances of overlapping observed between the no-fault and faulty data. This overlap in the low-dimensional data suggests that the high-dimensional data is also likely to be overlapping, posing challenges for the classifier in accurately classifying the correct class. Furthermore, when using the residuals as inputs the no-fault data and the faulty data are more clustered together compared to using the measurement signals. However, there is still some overlapping between the cases.

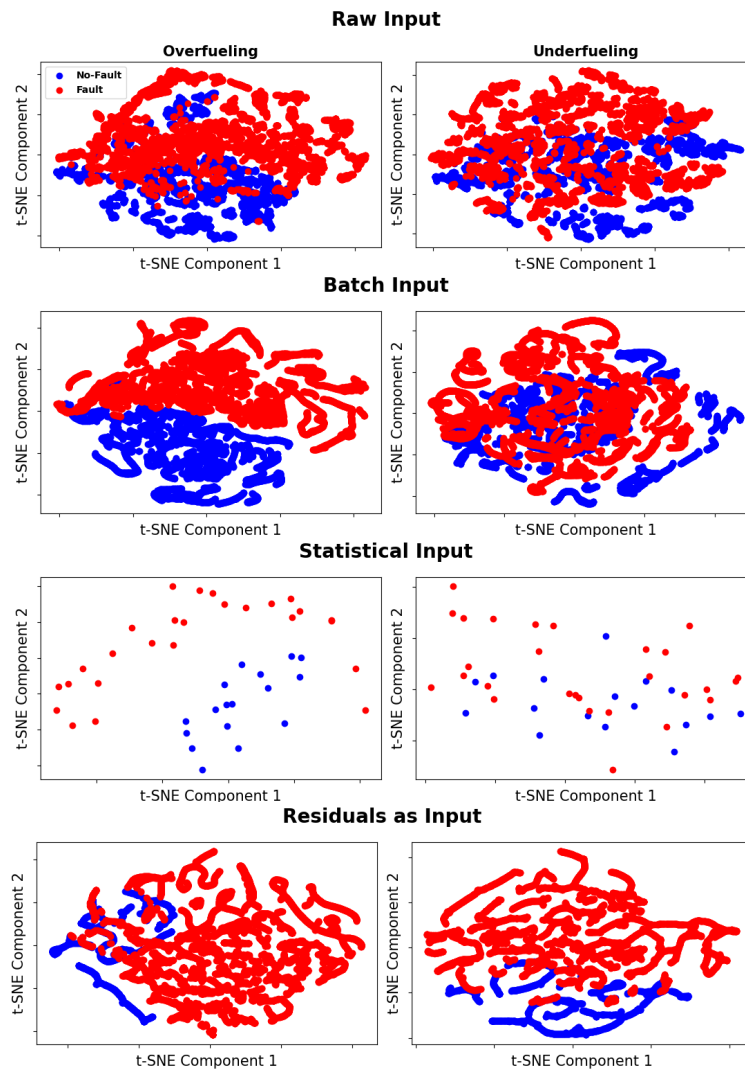


Figure 6.7: The result of the t-SNE analysis for the different inputs to the classifier during overfueling to the left and during underfueling to the right.

6.3 Operating Point Assessment

This section presents the results from assessing the impact of varying operating points on the different diagnostic methods.

6.3.1 Data-Driven Residuals

The histogram presented in Figure 6.8 illustrates the distribution of the NARX-based NO_x residual during the two no-fault cases (NF and NF Load) and the fault during the aggressive driving condition. While the histogram only displays the results of the NO_x residual, similar distributions are observed for the O_2 and exhaust pressure residuals and could be represented accordingly. As observed, all three scenarios are distributed around zero and are overlapping within an interval. However, there are data points belonging to the aggressive driving condition that reach values closer to 2, indicating that the CUSUM algorithm will detect this fault. It is worth noting that the residual during aggressive driving condition shows a much larger separation from NF compared to the normal driving condition in Figure 6.3.

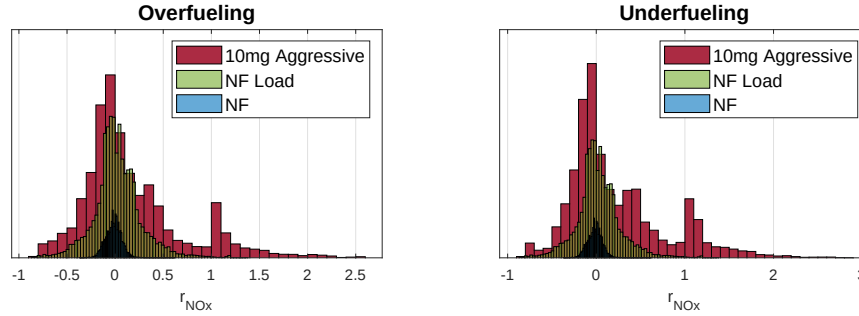


Figure 6.8: A histogram showcasing the distribution of the NO_x residual during different driving conditions. The NO_x residual is derived using the NARX model.

The CUSUM test quantity, $T(r_x)$, was used to assess whether the models can generalize across different operating areas, by comparing the output of the two no-fault cases. If they show similar values, it suggests that the model has good generalization. However, if the fault-free case with a load (NF Load) exhibits larger test quantity values, it indicates that the residual does not have values close to zero, which is a sign that the model is not predicting the output correctly.

The results of the CUSUM test for the aggressive driving condition and the NF Load are presented in Figure 6.9. Note that the results for NO_x were derived from the NARX-based residual rather than the SS, as the non-linear model demonstrated better generalization. The results for O_2 and P_{exh} were obtained using the SS-based residual. To achieve the NO_x and O_2 results, the design parameter ν had to be set higher than during normal driving conditions. This adjustment made the NO_x residual less sensitive to underfueling and the O_2 residual less sensitive to smaller faults in both underfueling and overfueling. The change in ν also caused the test quantity to increase during the NF Load condition ($T_{NF,Load}$), which is not a desirable outcome. Although the $T_{NF,Load}$ increased significantly, it is still possible to set a threshold that allows the diagnostic system to distinguish between no-fault and larger fault cases. When analyzing the $T_{NF,Load}$ for NO_x and O_2 , the typical values are approximately 60 with NO_x initially reaching slightly above 100. However, during aggressive driving and cases of larger faults from both overfueling and underfueling, the test quantity often reaches about 500 or higher. Even though T for a 15mg underfueling only reaches around 150, this

difference is considerable enough to suggest that the models might be robust against varying operating points. For P_{exh} , the value of the design parameter ν did not need to be adjusted from the normal driving condition, and the test output remained consistent. As previously observed, it effectively detects all faults except for 10mg underfueling. Additionally, the exhaust pressure model appears to generalize better than the NO_x and O_2 models, as the test quantity values show similar behavior across the two no-fault cases.

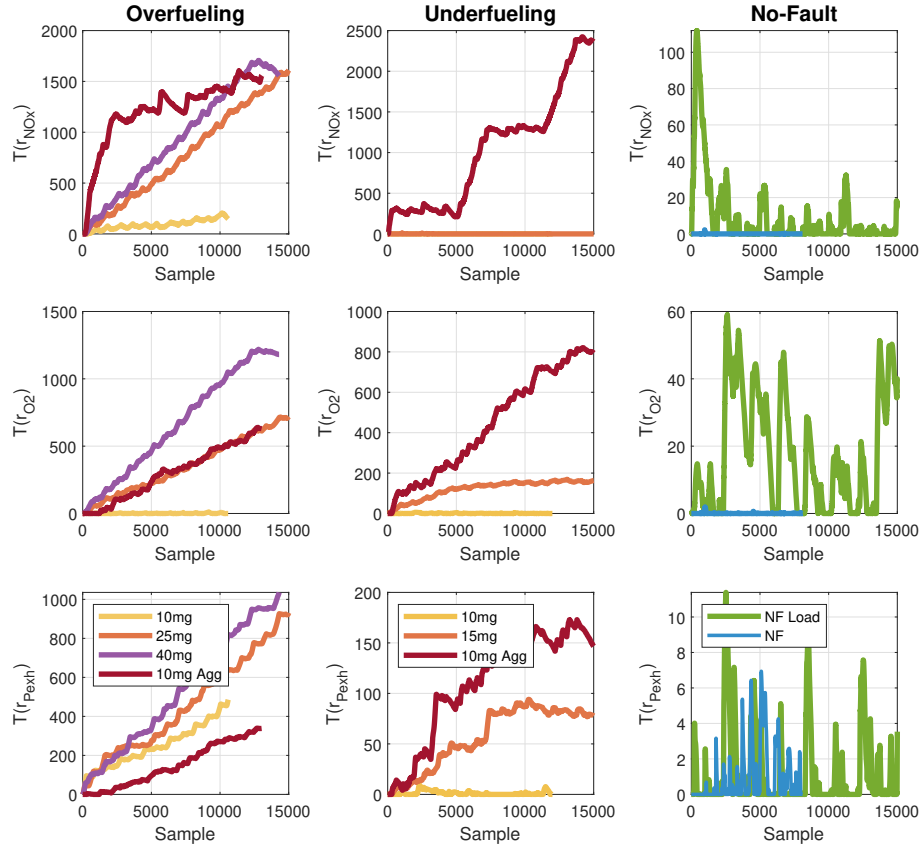


Figure 6.9: CUSUM test showing the NO_x , O_2 , and P_{exh} test quantities for normal driving and aggressive driving conditions, including various fault sizes for overfueling and underfueling. The two no-fault conditions are also shown. The test quantity for NO_x is derived from a NARX-based residual, while the test quantities for O_2 and P_{exh} are derived from SS-based residuals.

6.3.2 Random Forest Classifier

First, the results of the different input methods during overfueling will be presented, followed by the underfueling case. Then, the results of the classifier when exposed to and retrained on the data representing the different operating points will be presented.

6.3.2.1 Overfueling

In Figure 6.10, the resulting predicted probabilities for the different methods are shown. These plots are segmented by a dashed black line, where the left side represents the no-fault case with load, and the right side represents the 10 mg overfueling case with load and aggressive driving. As before, class 0 represents no-fault, and class 1 represents faulty. Both the raw

input and batch input show similar behavior. The model confidently classifies the no-fault case but struggles to differentiate the faulty case from the no-fault case, resulting in poor performance on the faulty data with many missed detections. The statistical input shows slightly better performance on the new data, being able to differentiate between the no-fault and faulty cases. However, as seen in the plot, the model's confidence in classifying the two different cases was not very high. Resulting in predicted probabilities hovering around the threshold, yielding both false alarms and missed detections. Feeding the model with the residuals as input did make the model more sensitive compared to using the measurement signals, but it also resulted in more false alarms.

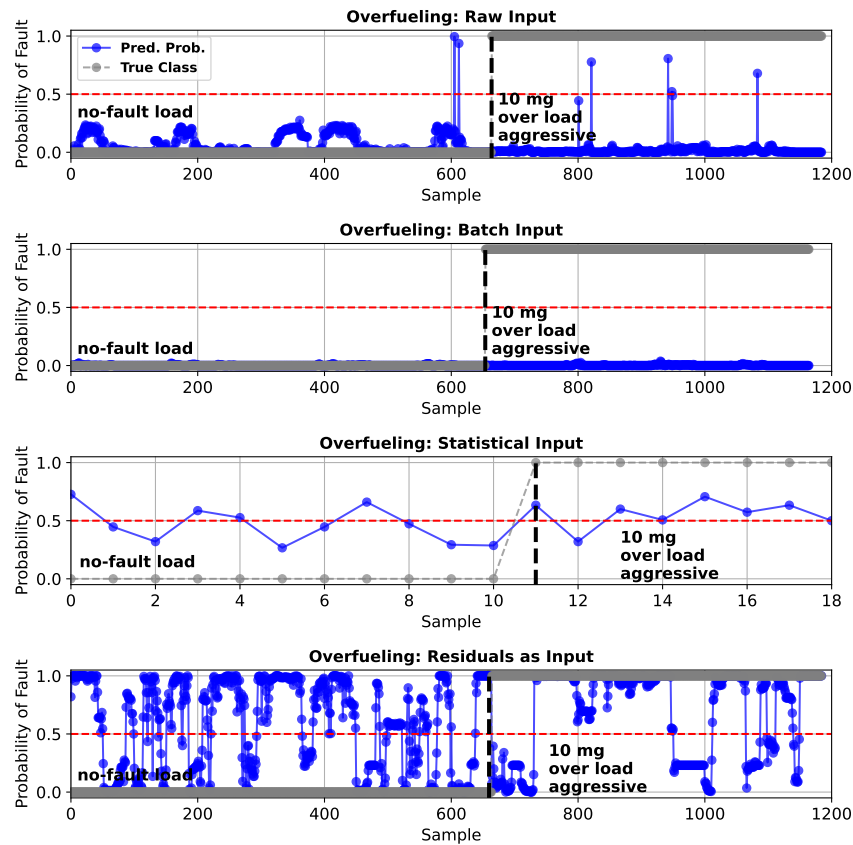


Figure 6.10: The predicted probability for the different inputs to the classifier during overfueling when fed with the new data. The red line indicates the threshold.

The performance for the different methods when fed with the new data is summarized in Table 6.4. It can be seen that the performance is relatively low for all methods, with the statistical input performing the best.

Table 6.4: F1-score and Accuracy for the different input methods during overfueling when fed with the new data.

Input:	Raw	Batch	Statistical	Residual
F1-score for No-fault (0.0):	0.72	0.72	0.70	0.46
F1-score for Fault (1.0):	0.01	0.00	0.67	0.54
Weighted Avg:	0.41	0.40	0.69	0.49
Accuracy:	0.5607	0.5611	0.6916	0.5012

6.3.2.2 Underfueling

The predicted probabilities during underfueling when the model was fed with the new data, can be seen in Figure 6.11. Similar to the overfueling case, the plots are divided by a black dashed line, now to the right representing 10 mg underfueling with load and aggressive driving. As seen in the plots, the model struggles to generalize on the new data for the underfueling case as well. It can be seen that the different models behave similarly during underfueling as they did in the overfueling case. The raw and batch input still struggles with the faulty case. It can be seen that the model yields some false alarms which it did not do for the overfueling case. The statistical input hovers around the threshold for the no-fault scenario, similar to what it did in the overfueling case. However, in the faulty scenario, the performance is worse and the model struggles to classify it correctly. Using residuals as inputs made the model more sensitive but also resulted in more false alarms, similar to overfueling. It performs better in classifying the no-fault condition compared to the faulty condition.

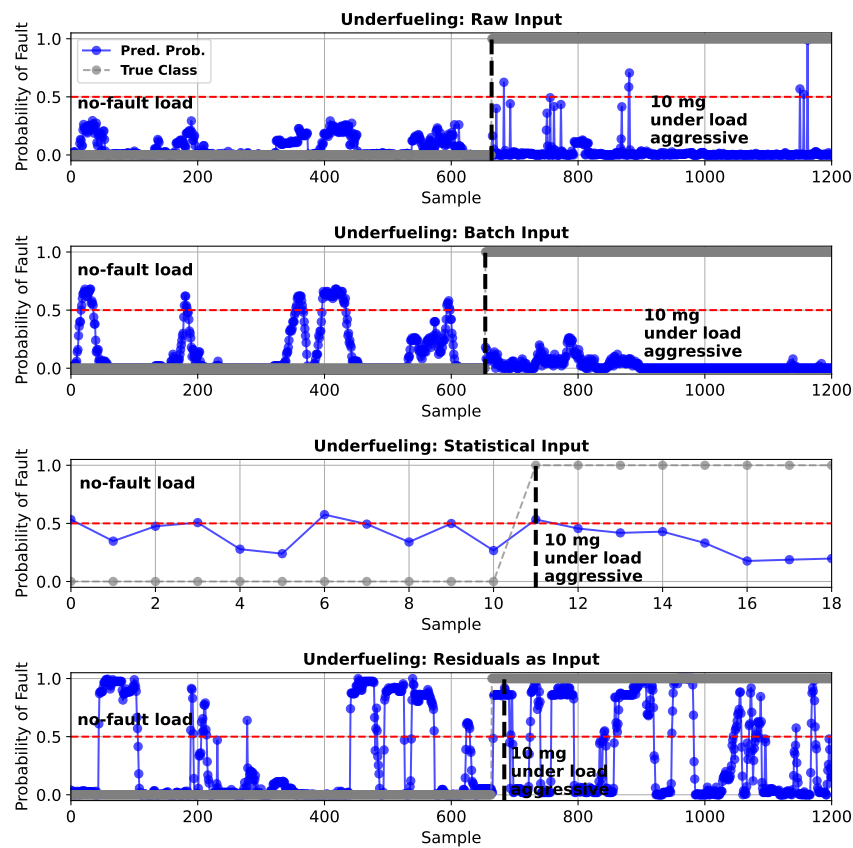


Figure 6.11: The predicted probability for the different inputs to the classifier during underfueling when fed with new data. The red line indicates the threshold.

In Table 6.5, the results for the different methods are summarized during underfueling. The overall accuracy is lower for the underfueling case compared to overfueling.

Table 6.5: F1-score and Accuracy for the different input methods during underfueling when fed with the new data.

Input:	Raw	Batch	Statistical	Residual
F1-score for No-fault (0.0):	0.68	0.68	0.54	0.62
F1-score for Fault (1.0):	0.02	0.00	0.13	0.47
Weighted Avg:	0.36	0.35	0.34	0.55
Accuracy:	0.5169	0.5129	0.3810	0.5574

6.3.2.3 Overfueling Retrained

The models were retrained on the training data, now also including the data representing the different operating conditions. They were then evaluated in the same way as before, by feeding the test data representing the different operating conditions. The result can be seen in Figure 6.12. The performance improvement is noticeable across all methods. Both the raw and batch input models now demonstrate sensitivity to the faulty case, performing well in classifying the faulty data. However, it is observed that these models yield more false alarms compared to before retraining them. The statistical input shows good performance after being trained on the new data, confidently distinguishing between no-fault and faulty conditions with only one false alarm and no missed detections. Using residuals as input also shows an increase in performance, although not as significant as the other methods. It still produces a considerable number of false alarms and some missed detections.

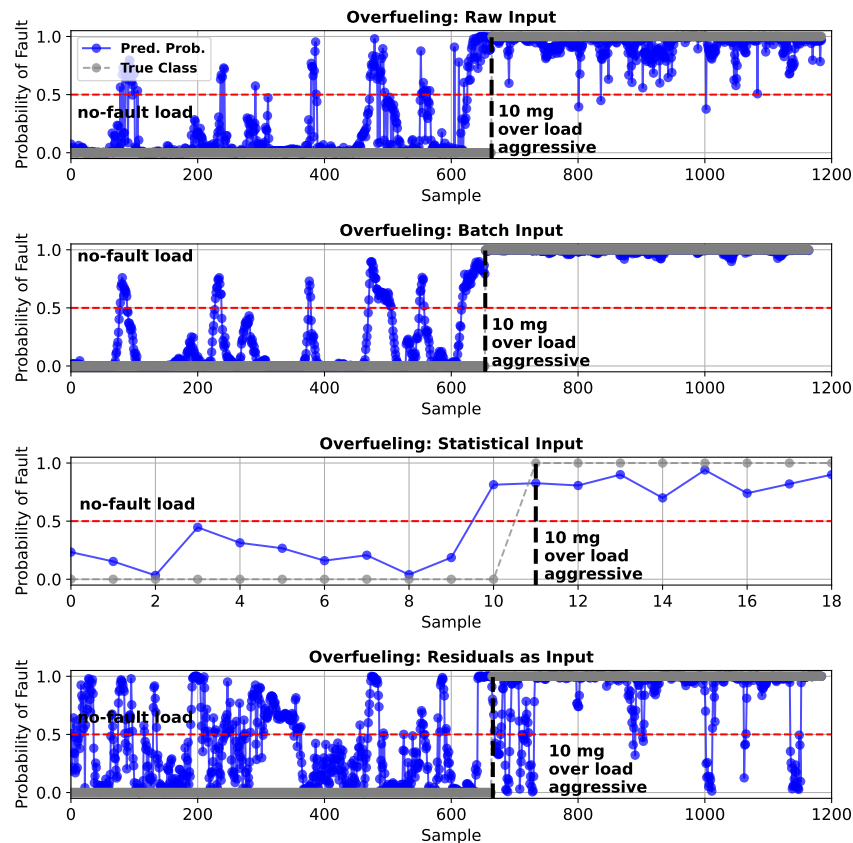


Figure 6.12: The predicted probability for the different inputs to the classifier during under-fueling when retrained on the new data. The red line indicates the threshold.

In Table 6.6, the performance for the different input methods is summarized. It can be seen that the raw, batch, and statistical inputs all achieved an accuracy above 90% after being trained on the new data. This is a significant improvement compared to their performance before training, where they achieved approximately 50% accuracy. This underscores the fact that the classifier's performance is heavily dependent on whether it has encountered the new operating point before. For the residuals as inputs, the improvement was not as big when retrained.

Table 6.6: F1-score and Accuracy for the different input methods during overfueling when retrained on the new data.

Input:	Raw	Batch	Statistical	Residual
F1-score for No-fault (0.0):	0.92	0.91	0.95	0.75
F1-score for Fault (1.0):	0.91	0.91	0.94	0.76
Weighted Avg:	0.91	0.91	0.94	0.75
Accuracy:	0.9121	0.9092	0.9474	0.7539

6.3.2.4 Underfueling Retrained

The same retraining procedure was performed for the underfueling case as for the overfueling case, and the results can be seen in Figure 6.13. Similar to the overfueling case, this led to a significant improvement in performance. In all plots, a dip towards the end of the fault case can be observed, which is probably due to noise in the data rather than the models themselves. Both the raw and batch inputs exhibit similar behavior, with an increase in false alarms but improved sensitivity in detecting the faulty case. The statistical input yields fewer false alarms and, except for the last part of the faulty section, shows good performance in detecting faulty behavior. The residuals as inputs show behavior similar to that observed in the overfueling case. Performance improves, but it still produces a considerable number of false alarms.

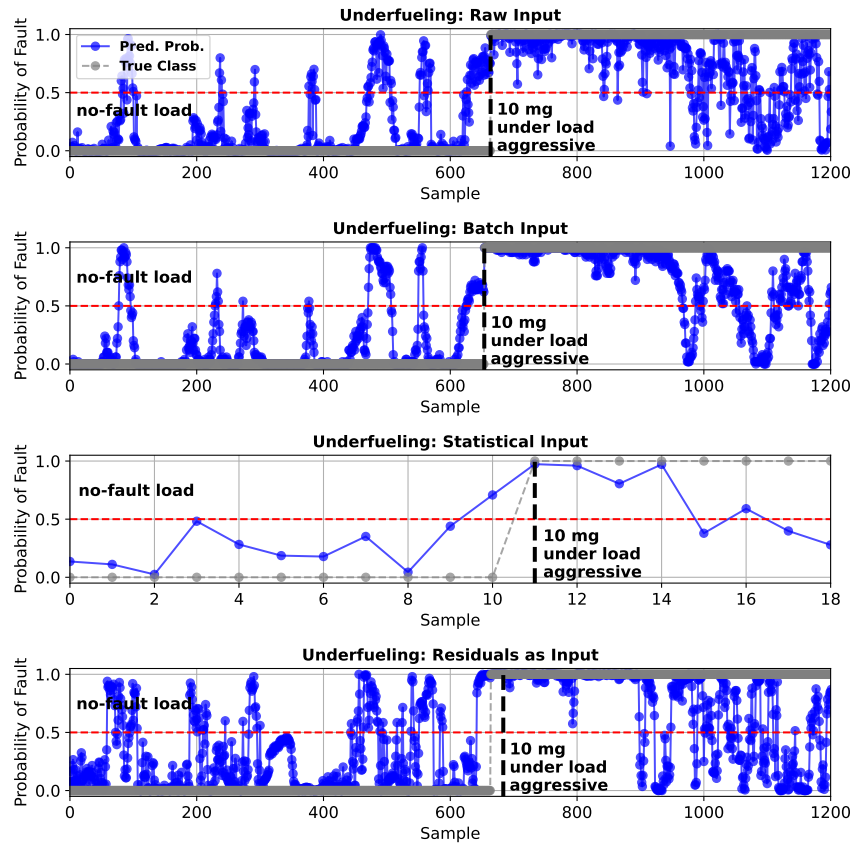


Figure 6.13: The predicted probability for the different inputs to the classifier during underfueling when retrained on the new data. The red line indicates the threshold.

In Table 6.7, the performance of the models is summarized. It can be observed that the performance for the underfueling case is slightly lower compared to the overfueling case. Overall, this result indicates that the classifier for the underfueling case is also very sensitive to having seen the different operating points before, for it to perform well.

Table 6.7: F1-score and Accuracy for the different input methods during underfueling when retrained on the new data.

Input:	Raw	Batch	Statistical	Residual
F1-score for No-fault (0.0):	0.81	0.82	0.80	0.73
F1-score for Fault (1.0):	0.79	0.80	0.71	0.69
Weighted Avg:	0.80	0.81	0.76	0.71
Accuracy:	0.8016	0.8097	0.7093	0.8016

Chapter 7

Discussion

This chapter begins by discussing the data collection process, highlighting the techniques and challenges encountered in gathering accurate and relevant data. Following this, a discussion of the results of the identification of informative target signals, the diagnosis methods employed, and the assessment of various operating conditions is presented. Finally, the chapter concludes with a comparison of the various methods utilized, presenting a proposed methodology and its practical implementation.

7.1 Data Collection

When collecting data, it was possible to gather overfueling data with a fault size of up to 40mg. However, during underfueling, the truck exhibited unusual behavior with larger fault sizes. Thus, it was decided to collect data only for fault sizes of 10mg and 15mg. It would be beneficial to collect data for larger underfueling faults to further investigate and understand how the system behaves. Ideally, having the same fault sizes for both overfueling and underfueling would allow for a better comparison of how the target signals are affected.

The reason for using a constant fault was that it provides a realistic representation of some of the observed fault modes [35]. An alternative approach could be to implement a fault that is proportional to the injected fuel. The signal for injected fuel mass is an estimated signal. Therefore, it may not accurately reflect the actual amount of injected fuel.

When collecting the data to validate the model's ability to generalize across different driving conditions, it is important to gather data that indicates whether or not the model is generalizing effectively. However, it is equally important to ensure that the training data is not unrepresentative to the point where the model does not stand a chance to generalize from the beginning. This goal may not have been fully achieved when collecting data under aggressive driving conditions. It might have been more effective to collect data by solely altering the driving style or maintaining a normal driving pattern with variable loads. When two variables are changed simultaneously, it becomes challenging to determine which factor influenced the evaluation results. It is also difficult to ascertain whether the change is too significant for the model to handle, based on its initial expectations. During data collection for the aggressive driving condition, the truck also carried a load. To more accurately represent the impact of the driving style, it would have been better to collect data without a load since the other fault data was gathered without one. As a result, the aggressive driving data reflects changes that are influenced by both the load and the driving style.

7.2 Identification of Informative Target Signals for Diagnosis of Injector Faults

This work initially took a broad approach in identifying informative target signals for diagnosing injectors causing over- and underfueling. To ensure relevance in the selected target signals, unreasonable and uninformative signals were removed based on their likelihood of being influenced by the fault. For instance, most temperature signals were removed because they were significantly affected by how the experiment was conducted. While some decisions involved subjectivity, this process narrowed the selection to the most informative signals for diagnosing this fault. This step was done before employing the two variable selection methods because those methods focus solely on the disparity between no-fault and faulty conditions. This could lead to high scores for irrelevant signals, despite differences originating from other known factors.

Both variable selection methods used in this work rely on the assumption that experiments for both the no-fault and faulty cases are conducted similarly, with only the fault distinguishing them. However, conducting experiments on a real truck poses significant challenges in ensuring this similarity. Additionally, both methods have their drawbacks. For instance, in absolute difference, the correlation between signals is not taken into consideration, and in FI in RF, highly correlated features may share importance, leading to ambiguity in determining which features are truly significant. This makes it unwise to directly trust that the signals with the highest scores are the most informative signals for diagnosing over- and underfueling. Instead, they should be seen as a secondary input grounded in experimental data to validate the signals identified in the literature.

7.3 Data-Driven Residuals

In this section, the results of the data-driven residuals are discussed.

7.3.1 Evaluation of Model Performance and Design

When comparing the performance of the SS, ARX, and NARX models, it was observed that they generally yielded similar predictions. However, the NARX model exhibited slightly superior results in most cases. Considering the system's inherent non-linearities, and especially the non-linearities in the discussed signals, these findings are explainable. Despite the system's non-linear behavior, the SS model still produced relatively accurate results, particularly for predicting O_2 concentration.

The selection of inputs influences the predictive accuracy of the models, therefore, it is desirable to have informative inputs. The selected inputs yielded promising predictive outcomes for the O_2 and P_{exh} , indicating that they were well-chosen. However, the NO_x prediction did not perform as well, achieving only around 35% accuracy. Achieving a more accurate prediction may necessitate selecting another set of inputs or incorporating additional ones. Regarding P_{exh} , various input combinations were evaluated. While several vehicle signals appeared to influence pressure, the decision was made not to utilize all of them, despite yielding a prediction with higher performance than that presented in Table 6.1. This decision was driven by the recognition that while including certain signals improved prediction accuracy, it reduced the sensitivity of the residual for fault detection. Therefore, it is worth noting that achieving a good prediction does not guarantee effective fault detection by the residual. This was also observed in the NO_x prediction, despite lower accuracy, the models still provided effective fault detection. Furthermore, although the NARX model offered more accurate predictions, diagnostic analysis suggests that the SS model is equally, if not more, effective in fault detection. In paper [11], researchers found similar results, with the

ARX model providing better predictions of fuel system pressure compared to the SS model. However, their research also revealed that the SS-based residuals exhibited higher sensitivity to faults. The two ARX models utilize past measurements to inform predictions, potentially allowing them to account for errors in new predictions during faults. In contrast, the SS model relies on internal state variables for predictions, which may lead to error accumulation over time. To conclude, if possible, opt for a simpler model such as SS, as it provides a reliable balance between simplicity and fault detection effectiveness.

7.3.2 Evaluation of Diagnosis and Residual Performance

Ideally, the parameter ν in the CUSUM algorithm (4.2) should have a value such that the test quantity during no-fault conditions, T_{NF} , is close to zero. Determining the optimal value of ν can be more challenging if the residual during no-fault is not symmetrically distributed. Setting ν too high might reduce the algorithm's sensitivity to faults, increasing the risk of missed detections, while setting it too low could lead to false alarms. However, it is not necessary for T_{NF} to remain zero for the CUSUM algorithm to be effective, as demonstrated in Figure 6.4. As long as T_{NF} does not exceed the threshold J and T_F (test quantity during a fault condition) exceeds J , the system can detect faults. Opting for a higher J might allow for a smaller value of ν , as it would tolerate larger values of T_{NF} without triggering an alarm, but at the expense of prolonging the fault detection time. Hence, finding the optimal balance between detection time and sensitivity is crucial.

It is logical that both the NO_x and exhaust pressure residuals can effectively detect over-fueling, given the significant impact the fault has on signal behavior as discussed in Section 7.2. In such cases, the deviation from nominal behavior is expected. However, the residuals struggle with detecting underfueling as this fault does not produce as pronounced effects on the signal behavior. Although they face challenges in detecting underfueling, they exhibit a slight increase of the CUSUM test quantity during 15mg, suggesting potential sensitivity to larger faults. In contrast, the O_2 residual successfully detected all faults and fault sizes with the CUSUM test in Figure 6.4. The residual's robust performance in detecting both over- and underfueling can be attributed to its responsiveness to faults. However, it showed a somewhat smaller response to the 10mg overfueling, with the test quantity T_F reaching only about 100. Nonetheless, because T_{NF} did not exceed 5, the O_2 residual could still detect the fault, but it would take slightly longer time compared to larger faults.

One challenge with the P_{exh} residual is its tendency to exhibit similar behavior in certain intervals in both no-fault and fault cases, often resulting in values close to zero. This similarity typically complicates distinguishing between the two cases. Nevertheless, the CUSUM test revealed that the P_{exh} residual was still able to detect overfueling. But, enhancing its robustness could potentially reduce detection time and enable it to also detect underfueling. One option to address this challenge is to implement criteria for determining when the residual should be applied. Additionally, since the residual during no-fault also exhibits larger deviations compared to the NO_x and O_2 residuals, determining an appropriate value for ν becomes more challenging. These deviations arise from inaccurately modeling nominal behavior under high-pressure conditions. This inaccuracy may be attributed to the lack of essential pressure dynamics in the model to comprehensively capture system behavior. Further investigation and refinement of the model could lead to improved prediction, ensuring that the no-fault residual remains within a narrower interval around zero. However, as discussed previously, model improvements may unintentionally reduce the sensitivity of the residual to detect faults. Despite this, improving the residual by minimizing deviations during no-fault periods may not be necessary, as the CUSUM algorithm can handle such deviations depending on the threshold.

7.4 Classification

In this section, the results of the classifier are discussed.

7.4.1 Evaluation of Performance and Input Method

The classifier achieved a test accuracy of approximately 98% for both over- and underfueling when fed with raw measurement signals. In study [8], ML algorithms were used for fault diagnosis in the fuel injection system and reached a test accuracy of 95%. Similarly, [15] conducted a study on engine fault diagnosis, specifically investigating injector faults among others, and achieved a test accuracy of 98.33%. Although their experiments were not identical to the ones in this project, as they used different algorithms and similar but not identical signals, these results indicate that the findings in this work are reasonable. The classifier performed better in distinguishing between no-fault and faulty conditions for overfueling compared to underfueling. This observation is consistent with the analysis of the t-SNE plots, where overfueling data points showed greater separation, indicating the same behavior in higher dimensions. However, comparing the separation between over- and underfueling may not be entirely fair, as the overfueling dataset includes instances with nearly double the fault size compared to underfueling. Nevertheless, when considering cases with the same fault size, overfueling had a more pronounced impact on the system. Instances of overlap between the no-fault and faulty states were also observed in the t-SNE plots, leading to situations where the classifier had to guess the correct class, resulting in false alarms and missed detections. The reason for this overlap is that the fault does not significantly impact the measurement signals to the extent required for clear separation. This underscores the need for further refinement of the classifier.

A strategy to address the issue of overlapping data involves preprocessing the data provided to the model to integrate additional or superior information without increasing the signal count. The decision was therefore made to try feeding the classifier with batches that include measurements from earlier time steps. This led to increased separation and decreased overlapping between the no-fault and faulty data, as can be seen in the t-SNE plots. The reduction in false alarms and missed detections, seen in the predicted probabilities, indicated a higher level of certainty in predicting the correct class for both over- and underfueling scenarios. However, this modification resulted in larger sets fed to the model, leading to longer training and prediction times, which was undesirable.

An alternative approach was therefore investigated, involving the calculation of the mean and standard deviation of segments of data. This method aimed to filter out noise and outliers, thereby aiming to reduce the overlapping. Additionally, by reducing the number of samples, the training and prediction times would decrease. The results of this method showed promising results especially for the overfueling case, as shown in the t-SNE plot and the predicted probability plot. However, for the underfueling case, the results were less promising. The model struggled to distinguish between the no-fault and faulty states, exhibiting greater confidence in classifying the faulty state compared to the no-fault state, which led to a high rate of false alarms.

Since the residuals contain similar information as the measurements, it was of interest to see how they would perform compared to the measurement signals. The performance was similar but slightly worse for the residuals. As observed in the t-SNE plots, the different cases were more clustered together, but with some overlapping data points. This slight decrease in performance can be attributed to the model's reliance on just three inputs for decision-making, as opposed to nine, even though the residuals carry the same information.

Additionally, as seen in Section 6.1, the residuals for the no-fault and faulty cases overlap, making it harder for the classifier to classify accurately.

7.4.2 Further Refinement for Utilizing Classifiers in Diagnostic Purposes

The methods mentioned above all showed limitations, highlighting the need for further refinement to develop an effective approach for detecting over- and underfueling using a classifier. One way to improve the performance of the classifier in this work would be to increase the threshold. In this work, false alarms carried a greater cost compared to missed detections. By increasing the threshold, the classifier would be less prone to false alarms. This adjustment would make the classifier less sensitive to faults, but as seen in the predicted probabilities, the classifier often struggled more with the no-fault state compared to the faulty state.

Another way to further refine this diagnostic approach would be to implement a sequential analysis approach, similar to a CUSUM algorithm. Instead of requiring a single prediction to trigger an alarm, it could be decided that, for example, the classifier, when fed with the mean and standard deviation, should detect four consecutive faulty states before issuing an alert. This approach would enhance robustness by considering a longer time period of system operation. Each segment spans 30 seconds, so requiring, e.g., four consecutive segments to indicate a fault would encompass two minutes of system activity before deciding the state of the system.

7.5 Operating Point Assessment

An important aspect to consider in the operating point assessment is that the fault was implemented as a constant magnitude, as illustrated in Figure 7.1. Since load and injected fuel are correlated, this resulted in a proportionally smaller impact of the fault as the load increased. Therefore, it may be more difficult to distinguish between the no-fault load case and the faults with load during aggressive driving.

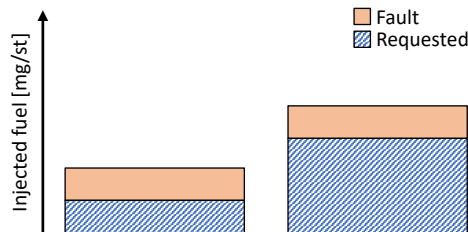


Figure 7.1: Illustration of the constant fault implementation in the project.

7.5.1 Data-Driven Residuals

Developing a model that can generalize across various operating conditions is challenging due to the complexity of a diesel engine, as reflected in the results for both NO_x and O_2 in Figure 6.9. Although $T_{NF,Load}$ (test quantity during NF Load) for these target signals displayed higher values, the CUSUM test indicates that the models can still differentiate between fault and no-fault conditions. That is because $T_{F,Agg}$ (test quantity during aggressive driving with fault) is significantly larger compared to $T_{NF,Load}$, making it possible to determine a reasonable threshold. However, these results required setting a higher value for the design parameter ν than during normal driving conditions. With the new value for ν , the $T_{NF,Load}$ for NO_x and O_2 reaches values between 40 and 60. Without this adjustment,

$T_{NF,Load}$ would exceed these levels, potentially triggering false alarms, as the residual tends to deviate further from zero in these scenarios compared to normal driving conditions. The downside of changing ν is that the models become less sensitive to smaller faults during normal driving. However, ν can be used as an adaptive threshold that adjusts based on the operating point, which can help avoid this problem. A higher ν is suitable for situations with greater model uncertainty, while a lower ν is appropriate for operating points with less model uncertainty [25]. Another alternative is to create models to predict the target signals for each operating point, with the choice of model depending on the current driving condition. Whether this approach is robust enough to use across various driving conditions requires further investigation, as it depends on factors like detection time and other specific requirements that are not defined in this project.

As mentioned in Section 6.3, only the NARX model was able to generalize effectively for the NO_x residual. This discrepancy is likely due to the complex non-linearities inherent in NO_x behavior, which the SS model struggles to capture across different operating points. Consequently, the SS model is not a viable option for this target signal when good generalization is required.

7.5.2 Classification

The classifier struggled to generalize on new data, particularly for the raw and batch input. In these cases, it often misclassified faulty data as no-fault, while correctly identifying instances of no-fault with load. This is reasonable because of how the fault was implemented. Since the fault remained constant at higher loads, its impact was proportionally smaller at higher loads, resulting in faulty data being clustered similarly to the no-fault data.

When the classifier was fed with the statistical input, sensitivity increased. However, as seen in the plots, the predicted probability for both the no-fault and faulty cases hovered around the threshold, indicating an uncertain model. This is likely due to the broader spread of data, which influenced the mean and standard deviation, making it harder for the classifier to accurately classify the correct class.

This showed that when encountering unseen data that does not neatly fit existing categories or exhibits shifts, the classifier struggled to perform as expected. However, when the classifier was trained using data representative of the new operating conditions, the performance improved significantly. This highlights that the classifier is data-hungry and needs to have a broad range of training data to perform well across different cases, which can be difficult and time-consuming to obtain.

When feeding the RF model using the residuals as inputs instead of the raw measurements, the classifier exhibited slightly better generalization compared to when fed with the measurements. This is reasonable because the residuals are not as spread out as the measurements. The underlying models capture the system dynamics, making them better at generalizing across different operating points. However, when retrained on the new data, it can be observed that the performance did not improve as much as with the measurements. This is likely because the residuals, unlike the new measurements, may not be as separated, leading to less noticeable distinctions.

7.6 Discussion of Diagnostic Methods

In this section, the methodology used in this study for fault detection is discussed and in Figure 7.2 the steps taken are outlined. This graphical overview provides insights into the design of the diagnosis methods investigated in this thesis, highlighting the limitations of the various methods and offering guidance on their appropriate application.

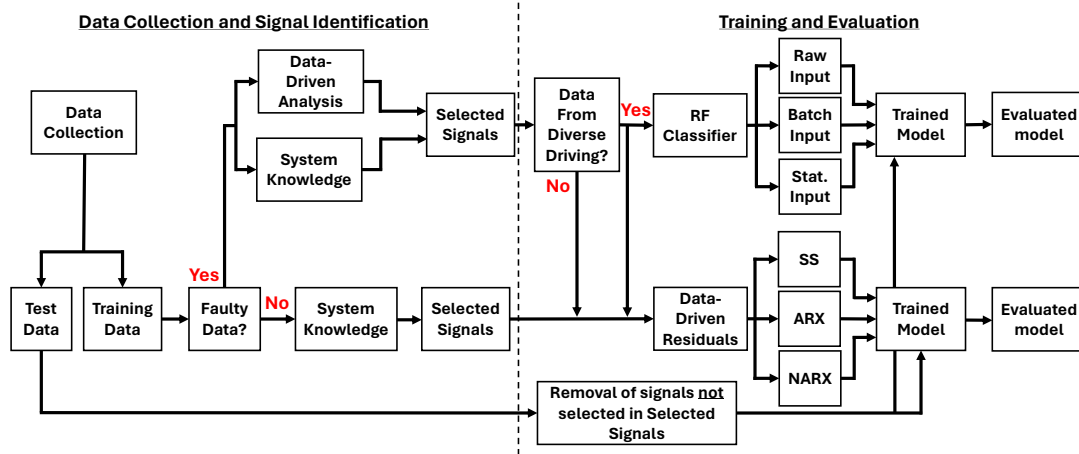


Figure 7.2: Graphical overview of the concepts evaluated in this work.

7.6.1 Data-Driven Residuals

Like other data-driven approaches, data-driven residuals require high-quality data for accurate training and validation. The data should encompass a wide range of operating conditions to ensure the models can generalize effectively. However, given that the models capture the system dynamics, it offer a more generalized approach compared to other methods. It is crucial to have a well-designed data collection plan to ensure high-quality data, as insufficient or poor-quality data can lead to unreliable models and inaccurate predictions. Gathering a comprehensive dataset can be challenging due to resource constraints, requiring time, equipment, and human expertise. Another limitation is variability among different vehicles. Even with the same make and model, slight differences in system components can lead to variations in system dynamics. This variability might hinder the creation of a universal model that performs reliably across different vehicles. Nonetheless, since the model captures underlying dynamics, it might still be generalizable across various vehicles, though additional research is needed to confirm this.

In data-driven approaches, model complexity tends to increase with the complexity of the system, potentially leading to computational challenges and longer training times. For instance, the complexity of SS models grows with the number of states and inputs, requiring more computational resources. However, as demonstrated in this thesis, simpler models effectively capture the dynamics of complex systems like diesel engines, thereby reducing computational costs. Another challenge is tuning these models, as developing accurate models may require a deep understanding of the system, which can be time-consuming. For instance, when designing NARX, custom regressors can be crafted based on insights into the system's non-linearities. Despite this, a data-driven approach can be a viable alternative to a model-based approach when system knowledge is limited. While some system knowledge is beneficial for designing data-driven models, it does not require as much detailed system information as model-based approaches.

Due to the NARX model's non-linear nature, these models may be more sensitive to small changes in input conditions, potentially affecting robustness and stability. One limitation of the SS model is that if it is not designed properly, it can lead to unstable behavior or oscillations. This requires careful tuning and validation to ensure stability and accurate fault detection. Despite these limitations, both SS and NARX models have their respective advantages, with SS offering simpler design for predominantly linear systems and NARX providing flexibility for more complex, non-linear systems.

7.6.2 Classification

With numerous supervised ML algorithms available, RF was chosen for several reasons. It is versatile, easy to use and understand, with minimal parameters to tune, making it a good choice. While there may be algorithms that outperform RF, it served well for evaluating the use of ML on this problem.

When utilizing a classifier for detecting over- and underfueling in a diesel engine, several considerations arise. While it is advantageous because there is no need for a model of the system, and its ability to handle non-linear relationships and adapt to new data, there are limitations. The limited interpretability makes understanding specific predictions challenging. The challenge of generalizing on new unseen data limits this method, and the need to see all different operating points can be both challenging and time-consuming. Furthermore, when the data is not well separated, the classifier performance is compromised as it resorts to guessing. This leads to both false alarms and missed detection. This can be problematic when dealing with this type of fault since the fault did not significantly differentiate the measurement signals between no-fault and faulty behavior. However, an enhancement was noticed in performance when preprocessing techniques were applied to the data. Therefore, if this method were to be implemented within a real system, preprocessing the data before feeding it to the classifier becomes imperative. A viable strategy would be to calculate the mean and standard deviation of data sets. This not only filters out noise but also reduces computational demands, which are limited in a truck. Instead of requiring predictions every tenth of a second, a prediction must be made every 30 seconds. To further enhance this preprocessing method, previous measurements could be incorporated into this method.

The method of feeding residuals to the classifier combines strategies from both the classifier and data-driven residuals, leveraging their strengths. This integration allows the models to capture the data dynamics and enables the classifier to make decisions based on all three inputs. However, as seen in the results, implementing this in a real system would require preprocessing the data similarly to measurement signals and it would also rely on two different methods working as intended, raising questions about its practicality.

7.6.3 Compilation and Comparison of the Diagnosis Methods

This work evaluated two methods for detecting over- and underfueling across different fault sizes. The first method involved data-driven residuals, which performed well in diagnosing both faults. Selecting the target signals to be predicted and the correct input signals for the models can be time-consuming and requires knowledge of the system. The next method, involving the RF classifier, also showed good performance, especially after preprocessing the data. Unlike the data-driven residuals, the RF methods do not demand extensive knowledge of the system, which can lead to faster development times. For instance, using the built-in feature importance can help select input signals, although it may not yield results as accurate as the data-driven residuals. Training the RF model requires faulty data, and it was observed that it does not generalize as well as the data-driven residuals unless the model was trained on a similar case before. It was seen that both methods require some form of sequential

analysis for robustness. The data-driven residuals provide greater interpretability, aiding in decision understanding. Combining both methods showed interesting results and can be an alternative if one wants to combine the better generalization capability of the residuals with the classifier to make decisions based on several inputs. Note that the residuals should also be processed before feeding them to the classifier to make this a viable option. If data from different scenarios is not available, data-driven residuals can be a good method to use. However, if data from many different cases is available and system knowledge is lacking, using an RF classifier can be an alternative. These findings are briefly summarized in Table 7.1.

Table 7.1: Short summarization of the two evaluated methods.

	Data-Driven Residuals	RF Classifier
Accuracy	Capable of detecting both faults and faults of all sizes.	Able to detect both faults and faults of all sizes, but data preprocessing is necessary.
Robustness	Requires the use of a CUSUM test, but shows good generalization ability.	Requires exposure to diverse scenarios and a sequential analysis is needed for robustness.
Development Time	Can take longer to determine which signal to predict and the correct input to the models. Does not require faulty data.	Is less sensitive to the selection of the correct input, but requires faulty data.
Interpretability	Offers high interpretability.	Interpretability can become challenging when dealing with numerous inputs.

Figure 7.3 illustrates a simplified version of how these methods would be implemented, highlighting their differences. However, this work did not take into account real-time demands in a truck. Further refinement would therefore be necessary for these methods to meet all requirements. Nevertheless, these methods proved to be promising concepts for detecting faulty injectors.

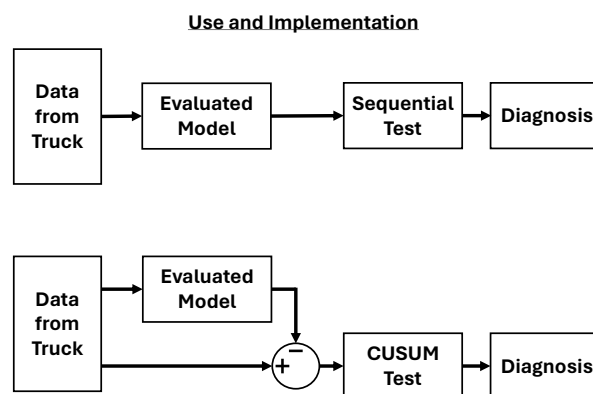


Figure 7.3: Graphical overview of the use and implementation would look like for the different methods. The upper illustration represents the RF classifier, while the lower represents the data-driven residuals.

7.7 The Work in a Wider Context

Improved fault detection in diesel engines offers benefits for the company such as cost savings, reduced downtime, and operational efficiency by providing a more efficient way of detecting engine faults. The improved detection may reduce the risk of accidents and equipment failures, enhancing the safety of employees and property. Environmentally, better diagnostics can lead to reduced emissions and a lower carbon footprint, aligning with sustainability goals. Ethical considerations are crucial in data-driven projects, emphasizing responsible data collection and resource use. Ensuring the model's robustness through extensive testing is challenging but essential. It is vital to cover all scenarios so that the model does not behave unpredictably when encountering unfamiliar situations.

Chapter 8

Conclusion and Future Work

This thesis work showed the exhaust gas emission signals NO_x , O_2 , and the exhaust pressure P_{exh} correlated with injector faults, suggesting their potential as additional signals for detecting over- and underfueling. However, this thesis only focused on injector faults. To further validate the effectiveness of these signals, it is necessary to explore how other types of faults influence these signals and determine whether the effects of over- and underfueling can be isolated from the other types of faults.

The combination of data-driven residuals and a CUSUM test quantity yielded promising results for fault detection in faulty injectors. The NO_x and P_{exh} residuals are demonstrated to be sensitive to overfueling, with some indication that they might detect larger faults related to underfueling. The O_2 residual, on the other hand, shows sensitivity to both overfueling and underfueling. The effectiveness of these residuals in terms of detection time, missed detections, and false alarms depends on the chosen design parameter ν and threshold, which vary based on specific requirements. The results also suggest that the non-linear NARX model provides the best signal prediction. However, a simpler linear SS model achieved better diagnostics performance. Although it does not offer the same level of prediction accuracy as NARX, the SS model still achieves good fault detection results, potentially even outperforming NARX in some cases, despite the system's non-linear characteristics.

The use of an RF classifier also yielded promising results for detecting over- and underfueling in all fault sizes. The method detected both faults but produced both false alarms and missed detections. This issue was due to the overlap of no-fault and faulty data, which hinders the effectiveness of the RF algorithm that requires distinct data separation. Feeding the classifier with the data-driven residuals did not improve the overlapping and the method showed similar performance. If this method were to be used, data preprocessing is essential. Both tested methods showed promising results. However, in a real system, using the statistical input is probably preferred as it shows good performance and less frequent prediction requirements. It could also be seen that if this method were to be used, tuning the threshold and using a sequential analysis would be good to improve robustness.

Data-driven residuals demonstrated adequate generalization across various operating conditions, though they did not fully generalize. Despite this limitation, the residuals were effective enough to distinguish between no-fault and fault conditions. Adjustments to the design parameter were required for the NO_x and O_2 residuals. These adjustments affected the sensitivity to smaller faults, leading to a compromise in which smaller faults might not be detectable across different operating points. The P_{exh} residual exhibited better generalization and did not require any adjustments. The appropriate threshold will likely vary depending on the specific application, suggesting that additional tuning may be necessary. To ensure the method's applicability across different scenarios, further investigation into its robustness is needed. The RF classifier, however, struggled to generalize across different operating

points. Using the data-driven residuals as inputs yielded a bit better generalization but not as good as using the CUSUM test. The classifier's performance significantly improved when retrained on new data, indicating its need for diverse scenarios. Obtaining such diverse data can be challenging and time-consuming, therefore the use of data-driven residuals might be a more efficient way for detecting over- and underfueling.

8.1 Future Work

Throughout this work, it was assumed that the truck was not affected by any other faults. However, if implemented in a real truck, it would be prudent to consider that other faults besides under- and overfueling could impact the diagnosis methods. Therefore, it would be advisable to implement and test with other common faults to assess their effect on the diagnosis system. This would allow for an evaluation of how well these methods can reliably diagnose under- and overfueling faults. In this project, only a constant fault in all injectors was investigated. For future work, it would be beneficial to explore the model's performance on other types of faults of the injector or examine the impact of a single faulty injector.

Further investigations into how these methods generalize across various operating conditions would be beneficial. This would further help identify conditions where the method performs well and where it falls short, providing insights into when to rely on it. For example, scenarios such as highway driving could be considered. Additionally, evaluating the performance of these methods on different trucks would be prudent to understand how well they perform in diverse environments.

For a deeper understanding of the effectiveness of data-driven residuals in diagnostic tasks,, future studies should focus on refining the design of the CUSUM threshold. The threshold needs to be carefully calibrated to meet specific criteria for detection time, false alarm rates, and missed detections, which will guide the improvement and reliability of the diagnostic process.

To further explore the use of classifiers in diagnosing over- and underfueling, it would be wise to test alternative methods such as neural networks. While RF classifiers have demonstrated effectiveness, they also have their limitations. Investigating other methods could potentially enhance this diagnostic approach.

Bibliography

- [1] T. V. Johnson. "Review of Diesel Emissions and Control". In: *SAE Int. J. Fuels Lubr.* 3 (2010).
- [2] JD. Gill, RL. Reuben, and J. Steel. "A study of small HSDI diesel engine fuel injection equipment faults using acoustic emission". In: *Journal of Acoustic Emission(USA)* 18 (2000).
- [3] S. Clever and HC. Isermann. "Model-based fault detection and diagnosis for common-rail injection systems". In: *MTZ worldwide* (2010).
- [4] D. Nyman. *Injector diagnosis based on engine angular velocity pulse pattern recognition*. 2020.
- [5] J. Kowalski. "Concept of the multidimensional diagnostic tool based on exhaust gas composition for marine engines". In: *Applied Energy* (2015).
- [6] R. Isermann. "Model-based Diagnosis of Diesel Engines". In: *MTZ worldwide* (2017).
- [7] B. Kleman and H. Lindgren. *Evaluation of model-based fault diagnosis combining physical insights and neural networks applied to an exhaust gas treatment system case study*. 2021.
- [8] Q. Shi, Y. Hu, and G. Yan. "A Novel Fault Diagnosis Algorithm for the Fuel Injection System of Marine Two-Stroke Diesel Engine". In: *2022 7th International Conference on Power and Renewable Energy (ICPRE)*. IEEE. 2022.
- [9] M. Fernandes, J. M. Corchado, and G. Marreiros. "Machine learning techniques applied to mechanical fault diagnosis and fault prognosis in the context of real industrial manufacturing use-cases: a systematic literature review". In: *Applied Intelligence* (2022).
- [10] D. Jung and C. Sundström. "A Combined Data-Driven and Model-Based Residual Selection Algorithm for Fault Detection and Isolation". In: *IEEE Transactions on Control Systems Technology* (2019).
- [11] N. Allansson, Mohammadi A., D. Jung, and M. Krysander. "Fuel injection fault diagnosis using structural analysis and data-driven residuals". In: *Journal Name* (2024).
- [12] E. Frisk, M. Krysander, and D. Jung. "A Toolbox for Analysis and Design of Model Based Diagnosis Systems for Large Scale Models". In: *IFAC World Congress*. Toulouse, France, 2017.
- [13] F. Kimmich, A. Schwarte, and R. Isermann. "Fault detection for modern Diesel engines using signal-and process model-based methods". In: *Control engineering practice* (2005).
- [14] O. Holmer and L. Eriksson. "Modeling and Analytical Solutions for Optimal Heating of Aftertreatment Systems". In: *IFAC-PapersOnLine* (2019).
- [15] C. Xie, Y. Wang, J. MacIntyre, M. Sheikh, and M. Elkady. "Using Sensors Data and Emissions Information to Diagnose Engine's Faults." In: *Int. J. Comput. Intell. Syst.* (2018).
- [16] A. Hultgren. *Automatic detection of the fuel composition in a Diesel Engine: Identifying fuel composition in the fuel system of a combustion engine and optimising for computational complexity*. 2021.

- [17] T. R. Krogerus, M. P. Hyvönen, and K. J. Huhtala. "A survey of analysis, modeling, and diagnostics of diesel fuel injection systems". In: *Journal of Engineering for Gas Turbines and Power* (2016).
- [18] M. A. Fayad. "Effect of fuel injection strategy on combustion performance and NOx/smoke trade-off under a range of operating conditions for a heavy-duty DI diesel engine". In: *SN Applied Sciences* (2019).
- [19] RC. Chen, C. Dewi, SW. Huang, and R. E. Caraka. "Selecting Critical Features for Data Classification Based on Machine Learning Methods". In: *Journal of Machine Learning Research* (2021).
- [20] O. Sagi and L. Rokach. "Ensemble learning: A survey". In: *WIREs Data Mining and Knowledge Discovery* (2018).
- [21] Y. Song and Y. Lu. "Decision tree methods: applications for classification and prediction". In: *Shanghai Archives of Psychiatry* (2015).
- [22] L. Breiman. "Random Forests". In: *Machine Learning* (2001).
- [23] B. H. Menze, B. M. Kelm, R. Masuch, U. Himmelreich, P. Bachert, W. Petrich, and F. A. Hamprecht. "A comparison of random forest and its Gini importance with standard chemometric methods for the feature selection and classification of spectral data". In: *BMC Bioinformatics* (2009).
- [24] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. "Scikit-learn: Machine Learning in Python". In: *Journal of Machine Learning Research* (2011).
- [25] M. Nyberg and E. Frisk. *Model Based Diagnosis of Technical Processes*. Linköping, Sweden: Linköping University, 2020.
- [26] L. Ljung and T. Glad. *Modellbygge och simulering*. Studentlitteratur AB, 2004.
- [27] MathWorks. *System Identification Toolbox*. R2022b. URL: <https://se.mathworks.com/products/sysid.html>.
- [28] G. Alcan, M. Unel, V. Aran, M. Yilmaz, C. Gurel, and K. Koprubasi. "Predicting NOx emissions in diesel engines via sigmoid NARX models using a new experiment design for combustion identification". In: *Measurement* (2019).
- [29] E. Fantenberg. "Estimation of Air Mass Flow in Engines with Variable Valve Timing". In: (2018).
- [30] L. Eriksson and L. Nielsen. *Modeling and Control of Engines and Drivelines*. John Wiley & Sons, Incorporated, 2014.
- [31] BS. Yang, X. Di, and T. Han. "Random forests classifier for machine fault diagnosis". In: *Journal of Mechanical Science and Technology* (2008).
- [32] L. van der Maaten and G. Hinton. "Visualizing data using t-SNE". In: *Journal of Machine Learning Research* (2008).
- [33] CE. Metz. "Basic principles of ROC analysis". In: *Semin Nucl Med* (1978).
- [34] A. A. Taha. "Metrics for evaluating 3D medical image segmentation: analysis, selection, and tool". In: *BMC Medical Imaging* (2015).
- [35] M. Lindblom. "Scania Group". 2024-03-01.